

BALANCE–FIELD FRAMEWORK (BFG)

The BFG Compact Canonical Universal Reclosure Equation — Freedom-Closed Unified V4

*A Fully Equivalent Minimal Operator Form of the Endogenous Dual-Order Strong
Universal Architecture with Intrinsic Persistence, Reciprocal Order, Polar Reclosure,
and No-Choice Natural Reduction*

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Scientific

status

Independent theoretical whitepaper and falsifiable research program. Unified V4 is an exact compact reformulation of the Freedom-Closed Unified V3 architecture. Mathematical universality is claimed only over the declared admissible BFG category; compactness introduces no new physical or empirical assumption. Natural-theory recovery and empirical Level-5 realization remain independently testable.

Abstract

This whitepaper develops the Compact Canonical Unified V4 form of the Balance-Field Framework (BFG) universal reclosure architecture. The aim is compression without loss: the Freedom-Closed Unified V3 generator, its exact neutral resolvents, recursive persistence, reciprocal dual order, polar support transport, positive Gram rebuild, formation gate, totalization, and natural-reduction discipline are retained unchanged, while their generator core is rewritten in a minimal operator form. From the positive master load $Y_X = K[X]^\dagger K[X] \geq 0$, define the derived neutral contrast operator $Z_X = (I - Y_X)(I + Y_X)^{-1} = C_N(Y_X) - B_N(Y_X)$. The original neutral pair is recovered exactly as $C_N = (I + Z_X)/2$ and $B_N = (I - Z_X)/2$. Hence the complete persistent reciprocal split is represented by a single compact operator $S_X = (1/2)\Omega_X[P_X(I + Z_X); P_X(I - Z_X)]$, where P_X is the canonical positive-metric persistent projector and Ω_X is the uniquely determined reciprocal-order weight operator. All downstream steps already fixed in Unified V3 are collected, but not replaced, by the canonical reclosure functional $\mathfrak{R}_X$: polar support transport, capacity compression, positive Gram reconstruction, exact admissibility gate, and final neutral reclosure. The universal equation is therefore $U[X] = \mathfrak{R}_X(S_X D[X])$, with $X_{\{n+1\}} = U[X_n]$, $U(\perp) = \perp$, and $X_n = U^n[X_0]$. An exact equivalence theorem shows that this compact form reproduces the expanded EDO-URE identically and introduces no new free parameter, approximation, physical postulate, carrier law, or empirical assumption. The contrast spectrum is automatically bounded by $-1 < Z_X \leq I$ for nonnegative Y_X , placing retention/complement balance on a universal normalized operator scale while preserving the original BFG neutral partition. The remainder of the paper retains the full Freedom-Closed architecture, natural-realization program, falsification protocol, numerical demonstrations, and

five-level natural-recovery certificate. Compact Unified V4 is therefore a notational and structural strengthening of Unified V3, not a replacement of its mathematical content.

Keywords: Balance-Field Framework; BFG; Compact Canonical Unified V4; universal reclosure; compact universal equation; neutral contrast operator; endogenous dual order; reciprocal balance; strong universality; Level-0; neutral resolvent; positive master geometry; spectral persistence; polar decomposition; canonical capacity transport; recursive persistence; freedom-closed reduction; no-retuning; intrinsic behavioral quotient; natural realization; consciousness; qualia

Contribution and source-line provenance

The Strong Universal Form, totalized admissible state category, Freedom-Closed theory-reduction layer, endogenous dual-order construction, and canonical polar transport are retained from the immediately preceding unified development. Marcel Theodor Wendé integrates here the Compact Canonical Unified V4 representation. The new compact form does not add a sector law: it derives a single neutral contrast operator from the existing positive master load, reconstructs the exact C_N/B_N pair by algebraic identity, and packages the already fixed downstream canonical operations into one reclosure functional. Earlier co-authored source lines remain credited numerically where their mathematics is used; the compact-equivalence statement and contrast-operator representation are new mathematical reformulations of the existing BFG architecture.

- Marcel Theodor Wendé and Thomas W. Loker: co-authored Level-0 foundation source line used for pre-polar admissibility, determinate formation, and the Level-0 boundary [3].
- Marcel Theodor Wendé and Luke Kenneth Casson Leighton: recursive conjugate-closure source line, including the form $R_C = MRM^\dagger$ with $M = \sqrt{N}$ used in the persistence architecture [4].

Reference and numbering convention

This whitepaper uses a single numerical reference sequence in order of first substantive appearance, with bracketed in-text citations and one numbered bibliography entry per source. Equations are numbered globally and consecutively. Figures and tables are numbered independently. Internal BFG documents are named only in the numbered bibliography; the body text uses source-line descriptions and numerical citations.

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1. Purpose, scope, and the unification criterion

The target is mathematical universality in a precise sense: one closed operator architecture must regenerate the structural data needed for its own next application without importing a stage label, a stage-specific order law, a tunable persistence selector, or a freely chosen next-level synthesis map. The criterion is therefore stronger than repeated use of

the same notation. Universality requires a self-map on one declared admissible category, canonical internal construction of every generator-level coordinate, an explicit terminal state for failed generation, and target-independent invariant classes. Empirical universality is a separate realization claim and is tested only after the mathematical classes have been constructed.

Unified V4 preserves every Freedom-Closed constraint of Unified V3 and adds one further objective: minimal expression without hidden compression freedom. The expanded generator remains the audit form, while the compact generator becomes the canonical display form. No new primitive operator is introduced. The neutral contrast operator is algebraically derived from Y, the split operator is exactly the existing persistent reciprocal neutral pair, and the reclosure functional is defined as the ordered composition of already canonical V3 steps. Compression is accepted only where exact expandability back to the V3 equations is proven.

$$\mathcal{T}_0 \rightarrow \Delta N_0 \rightarrow (P_1, N, P_2) \rightarrow K_0 \rightarrow \delta^* \rightarrow H_C \rightarrow Y_C \rightarrow \{C_N, B_N\} \rightarrow R_C \rightarrow \mathbb{U} \rightarrow \mathbb{U}^2 \rightarrow \dots \quad (1)$$

Equation (1) remains the red thread, but the iterated engine is now $\mathbb{U}$ rather than a recursion plus an externally supplied scalar order coordinate. The map is completed below as a totalized, level-independent self-map whose output carries the same admissible structural data needed for the next iteration. Named natural regimes are downstream realization hypotheses, never terms in the generator.

Unification

criticism

An alleged stage transition is not counted as unified if it requires a new sector-specific fundamental response function, a post-hoc role redefinition, a target-aware threshold, or a carrier map chosen only after the higher-level outcome is known.

2. Epistemic discipline and claim-state hierarchy

The claim-state hierarchy is retained rigorously [1,2]. Definitions specify the state category and equivalence relation; inherited structural axioms supply Level-0, exact neutral resolution, positive closure geometry, and recursive persistence; the Strong Universal completion specifies how these structures are transported under reclosure; the mathematical theorem is conditional on those declared axioms; physical principles such as local physical closure remain separate; and all identifications with natural systems remain empirical hypotheses.

Table 1. Claim-state discipline used throughout the paper.

Class	Meaning
Definition	Introduces an object or equivalence class; true only by stipulation within the formalism.
Axiom / inherited BFG structure	A structural assumption imported from the canonical corpus.
Theorem	Exact mathematical consequence under explicitly stated assumptions.
Conditional theorem	Exact consequence after additional assumptions not yet derived upstream.
Physical principle	A universal physical bridge constrained by no-retuning; LPCP is the canonical example [1].
Unification construction	New in this paper: a fixed self-reclosing operator recursion, including endogenous dual-order feedback, built only from inherited canonical objects and explicitly declared universal constructions.
Model completion	A concrete representational or toy choice used to make a calculation executable.

Class	Meaning
Spectral prescription	A fixed readout from the master spectrum; numerical content must be computed.
Empirical hypothesis	Identification of a formal closure class with an observed physical or phenomenal regime.
Synthetic simulation	Numerically exact for the stated toy model; not empirical validation.

The strongest formal conclusion is therefore not that every natural object has already been derived. It is that, once the Strong Universal admissible category is fixed, the same target-independent map closes on that category and generates only domain-independent invariant classes. The separate burden is to show that atoms, cells, organisms, conscious states, or other natural systems realize those classes under prospectively fixed measurement maps.

3. Canonical source architecture

3.1 Level-0 trinitarian source

The canonical Level-0 formulation defines a finite-dimensional real Hilbert space V_0 with three primitive admissibility roles: difference capacity $\mathfrak{D}_0$, coherence capacity $\mathfrak{c}_0$, and neutral admissibility $\mathfrak{N}_0$ [1,3].

$$\mathcal{T}_0 = (\mathfrak{D}_0, \mathfrak{c}_0, \mathfrak{N}_0) \quad (2)$$

$$K_0 = \mathfrak{c}_0 + \mathfrak{N}_0 - \mathfrak{D}_0 \quad (3)$$

$$\mathcal{A}_{\text{Adm}}(\delta) = \frac{1}{2}\langle \delta, K_0 \delta \rangle + \frac{1}{4}\|\delta\|^4, \quad \delta \in V_0 \quad (4)$$

A nontrivial isolated formation branch requires a simple negative minimum eigenvalue of K_0 and a positive spectral gap.

$$\lambda_0 = \min \sigma(K_0) < 0, \quad \text{mult}(\lambda_0)=1 \quad (5)$$

$$\Delta_0 = \lambda_1 - \lambda_0 > 0 \quad (6)$$

$$\delta^*_{\pm} = \pm\sqrt{-\lambda_0}e_0 \quad (7)$$

The quartic term makes the admissibility functional coercive; the Rayleigh–Ritz argument isolates the lowest eigenvector. The two minima are polarity-related. This is a mathematical formation result inside the stated real finite-dimensional model, not an empirical cosmological timestamp [1–3].

3.2 Exact neutral elimination

Let A map a difference channel into a neutral mediation channel. The canonical closure formulation uses a quadratic neutral completion whose auxiliary neutral variable can be eliminated exactly [1].

$$\mathcal{A}_C(D, \nu) = \frac{1}{2}\|D - A^\dagger \nu\|^2 + \frac{1}{2}\|\nu\|^2 \quad (8)$$

$$Y = A^\dagger A \geq 0 \quad (9)$$

$$\nu^* = (I + AA^\dagger)^{-1}AD \quad (10)$$

$$C_N(Y) = (I + Y)^{-1} \quad (11)$$

$$B_N(Y) = Y(I + Y)^{-1} = I - C_N(Y) \quad (12)$$

$$C_N(Y) + B_N(Y) = I \quad (13)$$

$$0 < C_N(Y) \leq I, \quad 0 \leq B_N(Y) < I \quad (14)$$

$$\mathcal{A}_C^{\min}(D) = \frac{1}{2}\langle D, C_N(Y)D \rangle \quad (15)$$

$$\frac{1}{2}\|D\|^2 = \frac{1}{2}\langle D, C_N D \rangle + \frac{1}{2}\langle D, B_N D \rangle \quad (16)$$

Equations (9)–(16) are the central exact identities of the unification. They supply a bounded retained response, a bounded complementary response, and an exact quadratic partition. Standard positive-matrix analysis supports the spectral and perturbative facts used here [7,8].

3.3 Positive master closure geometry and spectral sectors

The canonical universal closure architecture lifts the local neutral pair into a positive master closure geometry [1].

$$B_C = D_{\text{cov}} c \quad (17)$$

$$H_C = B_C^\dagger W_N B_C \geq 0 \quad (18)$$

$$Y_C = L_C^\dagger H_C L_C \geq 0 \quad (19)$$

Sector identity is licensed only by upstream spectral gaps. If $\mathcal{S}_C$ is the relevant self-adjoint master sector operator and Δ_a are disjoint spectral windows, then:

$$P_a = \mathbb{1}_{\{\Delta_a\}}(\mathcal{S}_C), \quad P_a P_b = \delta_{ab} P_a, \quad \Sigma_a P_a = I \quad (20)$$

$$\beta_a = B_N^{\wedge\{1/2\}} P_a B_N^{\wedge\{1/2\}}, \quad \beta_a \geq 0, \quad \Sigma_a \beta_a = B_N \quad (21)$$

This positive partition is the exact no-double-counting backbone of later readouts. A claimed higher-stage channel that has no upstream gap or invariant distinction is not permitted to become a new fundamental BFG sector simply by naming it.

3.4 Recursive persistence and source-line continuity

The recursive conjugate-closure source line is preserved explicitly. Its conjugate form is used in the persistence architecture [4]. The canonical universal architecture uses a protected persistence direction plus a stable complement [1].

$$R_C = M U_\Omega C_E M^\dagger, \quad M = \sqrt{N_R} \quad (22)$$

$$R_C e_T = e_T \quad (23)$$

$$\|R_C|_{\{E_s\}}\|_L \leq r < 1 \quad (24)$$

$$P_T = \text{RieszProj}_{\{\lambda=1\}}(Dc_{\text{phys}}) \quad (25)$$

For $s = a e_T + s_s$, the stable component contracts at least geometrically, so the full trajectory remains power bounded. This motivates the carrier-neutral witness notion used below. The mathematical distinction between eigenvalue information and robust subspace information follows standard perturbation theory [7,8].

4. Why endogenous dual order is canonical rather than ad hoc

4.1 Structural origin of the dual pair

The dual order state arises only after the exact neutral pair has already produced two mathematically distinct but complementary persistent responses. BFG therefore does not begin by postulating two order parameters. It begins with one retained branch and one complementary upward branch generated by C_N and B_N . The order problem is then: how can one assign a normalized internal mediation state to these two branches without inserting a new scale, a new sector coefficient, or a self-reinforcing preference for the already dominant branch? The answer used here is reciprocal balance: each branch receives the weight required to equalize its order-weighted closure load with the other branch. This is the minimal negative-feedback completion of a two-pole balance geometry.

4.2 Canonical persistent geometry

Let Y_n be the positive master load of the current state. Its strictly positive graph metric G_n defines the norm in which persistent projection and closure loads are measured.

$$G_n := I + Y_n > 0, \quad \|z\|_{G_n}^2 := \langle z, G_n z \rangle \quad (26)$$

Persistence itself is not selected by choosing a convenient contour. The witness space is defined by a target-independent dynamical criterion: recursive trajectories must remain bounded and must not asymptotically disappear. In the finite-dimensional power-bounded case with semisimple peripheral spectrum this is exactly the direct sum of the unit-modulus eigenspaces.

$$\begin{aligned} \mathcal{W}_n &:= \text{cl span}\{z : \sup_{k \geq 0} \|R_{\{C,n\}}^k z\|_{G_n} < \infty, \limsup_{k \rightarrow \infty} \|R_{\{C,n\}}^k z\|_{G_n} > 0\} \\ &= \bigoplus_{|\lambda|=1} \ker(R_{\{C,n\}} - \lambda I) \quad (\text{finite-dimensional power-bounded case}) \end{aligned} \quad (27)$$

When the peripheral block is isolated from the strictly decaying complement by a positive spectral gap, its Riesz projection is unique. Any contour enclosing exactly that isolated set yields the same projector; the contour is therefore a computational representation of a previously defined subspace, not a tunable model parameter [7–10].

$$\begin{aligned} \gamma_n &:= 1 - \sup_{\lambda \in \sigma(R_{\{C,n\}}) \setminus \sigma_{\text{per}}} |\lambda| > 0, \\ P_n^R &:= (1/2\pi i) \oint_{\Gamma(\sigma_{\text{per}})} (zI - R_{\{C,n\}})^{-1} dz, \quad \mathcal{W}_n = \text{Ran}(P_n^R) \end{aligned} \quad (28)$$

If W_n is any full-column-rank basis of $\mathcal{W}_n$, the master metric gives the unique G_n -orthogonal projector. The result is basis independent, idempotent, G_n -self-adjoint, and contractive in the G_n norm.

$$\begin{aligned} P_{\{\mathcal{W}_n\}^{\{(G)\}}} &:= \\ &W_n(W_n^\dagger G_n W_n)^{-1} W_n^\dagger G_n, \\ (P_{\{\mathcal{W}_n\}^{\{(G)\}}})^2 &= P_{\{\mathcal{W}_n\}^{\{(G)\}}}, \\ (P_{\{\mathcal{W}_n\}^{\{(G)\}}})^\dagger G_n &= G_n P_{\{\mathcal{W}_n\}^{\{(G)\}}} \end{aligned} \quad (29)$$

4.3 Exact keep, upward, and export roles

The persistent projector acts on both exact neutral responses. Keep, upward, and export are therefore roles derived from one partition and one persistence criterion, not independent gates.

$$\begin{aligned}
 D_n^{\text{keep}} &= P_{\{\mathcal{W}_n\}^{\{(G)\}}} C_N(Y_n) D_n, & (30) \\
 D_n^{\uparrow} &= P_{\{\mathcal{W}_n\}^{\{(G)\}}} B_N(Y_n) D_n, \\
 D_n^{\text{out}} &= (I - P_{\{\mathcal{W}_n\}^{\{(G)\}}}) B_N(Y_n) D_n \\
 \Lambda_n^{\text{keep}} &:= \|D_n^{\text{keep}}\|_{G_n}^2, & (31) \\
 \Lambda_n^{\uparrow} &:= \|D_n^{\uparrow}\|_{G_n}^2, \\
 S_n &:= \Lambda_n^{\text{keep}} + \Lambda_n^{\uparrow}
 \end{aligned}$$

4.4 Reciprocal Balance Principle and uniqueness

The dual-order map is fixed by two equations rather than chosen from a convenient functional family. First, the two weights must be a normalized positive partition. Second, the order-weighted closure loads must be equal. The second condition is the operational balance statement: neither persistent pole is permitted to dominate the next closure merely because its present load is larger.

$$\begin{aligned}
 \omega_n^{\text{keep}} &\geq 0, \quad \omega_n^{\uparrow} \geq 0, \quad \omega_n^{\text{keep}} + \omega_n^{\uparrow} = 1, & (32) \\
 \omega_n^{\text{keep}} \Lambda_n^{\text{keep}} &= \omega_n^{\uparrow} \Lambda_n^{\uparrow}
 \end{aligned}$$

Proposition — Reciprocal-order uniqueness. For $\Lambda_n^{\text{keep}} > 0$ and $\Lambda_n^{\uparrow} > 0$, Equation (32) has exactly one solution on the one-simplex. Solving the two linear constraints gives the cross-fed pair below. Thus the reciprocal form is not an extra empirical fit and contains no exponent, gain, threshold, or sector-specific coefficient.

$$\begin{aligned}
 \omega_{\{U,n+1\}} &:= (\omega_{\{n+1\}}^{\text{keep}}, \omega_{\{n+1\}}^{\uparrow}) & (33) \\
 &= (\Lambda_n^{\uparrow}/S_n, \Lambda_n^{\text{keep}}/S_n) \in \Delta^1
 \end{aligned}$$

The pair has only one independent coordinate. Its projective odds recover a scalar balance coordinate with balanced value one. This gives a direct bridge to carrier-level scalar order without identifying the internal pair with any particular measured order variable.

$$\begin{aligned}
 r_{\{U,n+1\}} &:= \omega_{\{n+1\}}^{\text{keep}} / \omega_{\{n+1\}}^{\uparrow} & (34) \\
 &= \Lambda_n^{\uparrow} / \Lambda_n^{\text{keep}}, \quad \theta_{\{U,n+1\}} := \log r_{\{U,n+1\}}, \\
 \omega_{\{n+1\}}^{\text{keep}} &= (1 + e^{-\theta_{\{U,n+1\}}})^{-1}, \\
 \omega_{\{n+1\}}^{\uparrow} &= 1 - \omega_{\{n+1\}}^{\text{keep}}
 \end{aligned}$$

At $\Lambda^{\text{keep}} = \Lambda^{\uparrow}$, the unique state is $(1/2, 1/2)$ and $r_U = 1$. If one load grows relative to the other, its own order weight decreases while the opposite weight increases. The mechanism is therefore reciprocal negative feedback. A branch with zero closure load cannot support a genuine dual reclosure and is mapped to the terminal case by the universal gate below.

4.5 Cross-fed packet and exact non-expansion

$$\begin{aligned}
 \bar{D}_{\{n+1\}}^{\text{DO}} &:= \sqrt{(\omega_{\{n+1\}}^{\text{keep}} D_n^{\text{keep}} \oplus \sqrt{(\omega_{\{n+1\}}^{\uparrow} D_n^{\uparrow})})} & (35) \\
 \|\bar{D}_{\{n+1\}}^{\text{DO}}\|_{G_n \oplus G_n}^2 &= 2 \Lambda_n^{\text{keep}} \Lambda_n^{\uparrow} / S_n & (36) \\
 &\leq S_n / 2 \leq \|D_n\|_{G_n}^2 / 2
 \end{aligned}$$

Equation (36) is an exact local theorem, not a simulation result. C_N and B_N are commuting positive contractions with $C_N + B_N = I$, the witness projection is G_n -orthogonal, and the reciprocal weights solve Equation (32). Consequently the dual-order coupler cannot manufacture norm gain inside the structural master metric. This directly excludes an interpretation of endogenous order as a hidden amplification source.

5. Compact Canonical BFG Universal Reclosure Architecture

5.1 Compact Canonical Universal Equation

The expanded EDO-URE is retained as the audit form, but its neutral core can be compressed exactly. For a current admissible state X , let $D[X]$ denote its active carrier/difference object, let $Y_X = K[X]^\dagger K[X] \geq 0$ be the already canonical positive master load, let $P_X = P_{\{\mathcal{W}[X]\}}^\wedge(G_X)$ be the canonical positive-metric persistent projector, and let Ω_X be the diagonal reciprocal-order weight operator generated uniquely from the two persistent branch loads. Define the derived neutral contrast operator Z_X by the difference of the exact neutral resolvents.

$$Z_X := C_N(Y_X) - B_N(Y_X) = (I - Y_X)(I + Y_X)^{-1}, \quad Y_X := K[X]^\dagger K[X] \geq 0 \quad (\text{BFG-C1})$$

Because $C_N + B_N = I$, the two neutral channels are recovered exactly from Z_X . Thus Z_X is not a third neutral degree of freedom; it is a lossless one-operator encoding of the pre-existing pair.

$$C_N(Y_X) = \frac{1}{2}(I + Z_X), \quad B_N(Y_X) = \frac{1}{2}(I - Z_X) \quad (\text{BFG-C2})$$

The complete persistent reciprocal split can therefore be written as one state-dependent linear operator acting on $D[X]$. Here Ω_X acts on the two direct-sum components and contains no independent parameter.

$$S_X := \frac{1}{2} \Omega_X [P_X(I + Z_X) ; P_X(I - Z_X)], \quad \tilde{D}_X^{\wedge DO} = S_X D[X] \quad (\text{BFG-C3})$$

Let $\mathfrak{R}_X$ denote only the ordered composition already fixed by the expanded V3 construction after formation of $\tilde{D}_X^{\wedge DO}$: support polar decomposition of the cross-fed analysis operator, canonical capacity compression, positive Gram reconstruction, the exact universal formation gate, and the final neutral reclosure. $\mathfrak{R}_X$ is therefore a composition symbol, not an additional map that may be tuned independently. The compact universal equation is:

$$U[X] := \mathfrak{R}_X(S_X D[X]), \quad X_{\{n+1\}} = U[X_n], \quad U(\perp) = \perp, \quad X_n = U^n[X_0] \quad (\text{BFG-U})$$

This is the canonical display equation of Unified V4. The expanded equations below remain the definitional audit of $\mathfrak{R}_X$ and verify exactly how transport, positive rebuild, admissibility, and terminality are implemented.

Conditional Theorem — Compact/Expanded Equivalence. For every state X in the declared admissible BFG category for which the expanded V3 operators are defined, Equations (BFG-C1)–(BFG-C3) reproduce the original cross-fed analysis packet exactly. Since $\mathfrak{R}_X$ is defined as the same downstream composition used in the expanded EDO-URE, Equation (BFG-U) and the expanded EDO-URE yield the same next state or the same terminal state $\perp$. Therefore Unified V4 adds no new assumption, degree of freedom, approximation, carrier-specific law, or empirical identification.

The contrast representation also yields an immediate normalized spectral readout. If $y \in \sigma(Y_X)$ with $y \geq 0$, then $z = (1-y)/(1+y)$. Hence $-1 < z \leq 1$, with $z=1$ at zero load, $z=0$ at unit load, and $z \rightarrow -1$ for large load. This is a derived operator normalization of the same BFG neutral partition; it is not a replacement for the carrier-level scalar order readout Φ or φ .

$$z(y) = (1-y)/(1+y), \quad -1 < z(y) \leq 1 \quad \text{for } y \geq 0 \quad (\text{BFG-C4})$$

5.2 Expanded audit I: canonical cross-fed analysis operator

The remaining potential freedom is next-level synthesis. Instead of postulating a map J_n , the Canonical Closure Edition first defines the analysis operator generated by the already fixed neutral pair, persistent projector, and reciprocal order weights.

$$\begin{aligned} \mathcal{A}_n^{\{DO\}} &:= [\sqrt{(\omega_{n+1}^{\{keep\}})P_{\mathcal{W}_n}} \\ &\quad \wedge \{(G)\}C_N(Y_n) ; \\ &\quad \sqrt{(\omega_{n+1}^{\{1\}})P_{\mathcal{W}_n}} \wedge \{(G)\} \\ B_N(Y_n)], \quad D_{n+1}^{\{DO\}} &= \mathcal{A}_n^{\{DO\}} D_n \end{aligned} \quad (37)$$

Its polar decomposition determines a unique partial isometry on the active support. Let $Q_n = (\mathcal{A}_n^{\{DO\}} \mathcal{A}_n^{\{DO\}})^{\{1/2\}}$. The Moore–Penrose inverse is used only on $\text{supp}(Q_n)$. The resulting J_n is fixed by X_n and satisfies $J_n^\dagger J_n = P_{\text{supp}(Q_n)}$; on the active support it is an isometry. No target label or free transport matrix remains.

$$\begin{aligned} Q_n &:= (\mathcal{A}_n^{\{DO\}} \mathcal{A}_n^{\{DO\}})^{\{1/2\}}, \\ J_n &:= \mathcal{A}_n^{\{DO\}} Q_n^{\{+\}}, \\ J_n^\dagger J_n &= P_{\text{supp}(Q_n)} \end{aligned} \quad (38)$$

5.3 Expanded audit II: canonical capacity transport and positive rebuild

Every self-adjoint capacity operator is transported by compression through the same polar partial isometry. The new active carrier space is the closed range of J_n inside the direct-sum space, so the transported capacities and the cross-fed packet live on the same mathematical object. Positivity, where present, and self-adjointness are preserved by compression. The support-coordinate expression $J_n^\dagger (\mathfrak{A}_n \oplus \mathfrak{A}_n) J_n$ is unitarily equivalent to the range compression. The next formation operator is therefore a derived object rather than a stage-dependent choice.

$$\begin{aligned} \mathcal{H}_{n+1} &:= \text{cl Ran}(J_n), \quad P_{n+1} := J_n J_n^\dagger, \\ \mathfrak{A}_{n+1} &:= P_{n+1} (\mathfrak{A}_n \oplus \mathfrak{A}_n) P_{n+1} \\ \mathcal{H}_{n+1} &\simeq J_n^\dagger (\mathfrak{A}_n \oplus \mathfrak{A}_n) J_n, \\ \mathfrak{A} &\in \{\mathcal{D}, c, N\}, \\ K_{n+1} &:= c_{n+1} + N_{n+1} - \mathcal{D}_{n+1} \end{aligned} \quad (39)$$

The cross-fed packet then enters the same positive Gram reconstruction used by the universal closure architecture. Positivity is regenerated, not assumed inductively.

$$\begin{aligned} H_{n+1} &:= B_C[D_{n+1}^{\{DO\}}]^\dagger W_N[D_{n+1}^{\{DO\}}] B_C[D_{n+1}^{\{DO\}}] \geq 0, \\ Y_{n+1} &:= L_C[D_{n+1}^{\{DO\}}]^\dagger H_{n+1} \\ &\quad L_C[D_{n+1}^{\{DO\}}] \geq 0 \end{aligned} \quad (40)$$

5.4 Expanded audit III: universal gate, terminality, and the EDO-URE

A nonterminal branch must possess both persistent poles, an isolated witness gap, and the same simple-negative-mode formation geometry used at Level-0. Failure is never repaired by adding a sector-specific fallback rule.

$$\begin{aligned} g_U^{\{DO\}}(n) &:= \mathbb{1}_{\{ \Lambda_n^{\{keep\}} \Lambda_n^{\{1\}} > 0, \\ &\quad \nu_n > 0, \lambda_{\min}(K_{n+1}) < 0, \\ &\quad \text{mult}(\lambda_{\min}(K_{n+1})) = 1, \Delta_{n+1} > 0 \}} \end{aligned} \quad (41)$$

$$\begin{aligned} X_{n+1} &= U[X_n] := \{ C_N(Y_{n+1}) D_{n+1}^{\{DO\}}, \text{ if } g_U^{\{DO\}}(n) = 1; \\ &\quad \perp, \text{ if } g_U^{\{DO\}}(n) = 0 \} \end{aligned} \quad (42)$$

$$\begin{aligned} U: X_{\text{adm}} &\rightarrow X_{\text{adm}}, \\ X_{\text{adm}} &:= X_{\text{adm}} \cup \{\perp\}, \quad U(\perp) = \perp, \\ X_n &= U^n[X_0], \quad \omega_{U, n+1} \\ &= \mathcal{O}_U[U^n(X_0)] \end{aligned} \quad (43)$$

5.5 Carrier order is a readout, not a generator input

The earlier scalar order relation remains useful at the empirical interface. Its status is now unambiguous: a carrier-specific scalar may be compared prospectively with the internal projective coordinate r_U , but it does not enter $\mathbb{U}$. A successful carrier theory must specify this readout before outcome inspection.

$$\begin{aligned} \varphi_{\text{obs}} \nu(\varphi_{\text{obs}}) &= \Phi_{\text{src}}, & \varphi_{\text{obs}} &\approx \\ \mathcal{R}_{\varphi}[r_U(X)] & \quad (\text{carrier-level realization only}) \end{aligned} \tag{44}$$

5.6 Hilbert-space closure beyond finite matrices

The universal resolvent core does not depend on finite matrix dimension. Let K be a closed densely defined operator on a separable Hilbert space. Then $Y = K^\dagger K$ is nonnegative self-adjoint, so the spectral theorem gives bounded positive resolvents even when Y itself is unbounded [9,10]. This supplies the natural operator-theoretic extension needed for continuum carrier models without changing the neutral equations.

$$\begin{aligned} Y := K^\dagger K \geq 0 \text{ self-adjoint}, & \quad C_N(Y) := (I + Y)^{-1}, \\ B_N(Y) &:= I - C_N(Y), \\ 0 \leq C_N(Y) \leq I, & \quad 0 \leq B_N(Y) \leq I \end{aligned} \tag{45}$$

The corresponding positive master geometry is the quadratic-form graph space. All load and metric-projection statements are interpreted on its closed form domain. The finite-dimensional theory is the bounded special case.

$$\begin{aligned} \mathcal{Q}(Y) &:= \text{Dom}(Y^{\wedge\{1/2\}}), & \|z\|_{\mathcal{G}}^2 &:= \|z\|^2 + \|Y^{\wedge\{1/2\}}z\|^2, \\ & & z\|^2, \end{aligned} \tag{46}$$

G-form-closed witness subspaces admit unique
orthogonal projections

5.7 Compact canonicity and no-hidden-retuning theorem

Conditional theorem. Under the declared Level-0 formation assumptions, exact neutral elimination, the canonical bounded non-decay persistence definition, positive spectral isolation, reciprocal balance, polar support transport, positive Gram reconstruction, and the common exact formation gate, every generator-level object is fixed by X_n and the universal rules. The polar partial isometry uses the Moore-Penrose support convention and zero action on the kernel, so no arbitrary kernel extension is permitted. Degenerate or non-isolated determinate-formation cases are not resolved by a preferred vector or fitted tie-break parameter; they fail the declared determinate branch and map to $\perp$. Mathematical gate conditions are exact; numerical tolerances belong only to downstream measurement models. Thus $\mathbb{U}$ contains no stage predicate, external order coordinate, tunable persistence window, free kernel extension, level-specific synthesis map, or hidden numerical threshold. Unified V4 adds no generator object beyond these. Z_X is determined algebraically by Y_X , S_X is the exact persistent reciprocal neutral split, and $\mathcal{R}_X$ is only the named composition of the already unique downstream operations. Thus compactness introduces no new choice surface.

Figure 1. Canonical Endogenous Dual-Order Strong Universal Reclosure architecture. The order pair is uniquely fixed by reciprocal balance, the witness subspace is defined by bounded non-decay, and next-level transport is the polar partial isometry of the cross-fed analysis operator. The same engine rebuilds positive geometry and either returns the next admissible state or terminates at $\perp$.

6. Target-independent invariant classes and natural realization

The Endogenous Dual-Order Strong Universal Form removes named stage predicates from the generator. The Freedom-Closed edition strengthens this further: it does not permit an analyst to choose a different quotient for atoms, cells, gravity, quantum mechanics, or consciousness. All generator states are first reduced by one intrinsic future-behavioral equivalence relation defined solely from the complete sequence of canonical BFG data under repeated $\mathbb{U}$. Natural labels are applied only after that quotient exists.

$$Q_U := \bar{X}_{\text{adm}} / \equiv_U \quad (47)$$

$$\begin{aligned} \Pi_s: N_s &\rightarrow Q_U, \\ \varepsilon_s(x) &:= d_U(\Pi_s(T_s x), [U(X_x)]_{\equiv_U}) \end{aligned} \quad (48)$$

Natural names enter only through realization maps Π_s from independently described empirical systems into the universal quotient. A proposed atom, molecule, protocell, organism, conscious state, or qualitative content is therefore not generated because a predicate demanded that label. It is supported only if prospectively measured carrier data map into an already-existing invariant class and if observed transitions commute with $\mathbb{U}$ within the frozen error metric.

7. Physical closure from the same architecture

7.1 Local Physical Closure Principle

The Local Physical Closure Principle (LPCP) states that once an admissible closure enters a smooth, reversible, locally geometric regime, local propagation is the continuum limit of covariantly transported neighboring closure differences, using the same positive neutral stiffness that controls admissibility [1]. LPCP is a physical principle, not a Level-0 theorem.

$$\Delta_{\{\varepsilon, v\}} q(x) = U_C(x, x + \varepsilon v) q(x + \varepsilon v) - q(x) \quad (49)$$

$$\Delta_{\{\varepsilon, v\}} q / \varepsilon = \nabla_v \wedge C q + O(\varepsilon) \quad (50)$$

$$Z_C = H_C^{\{1/2\}} C_N(Y_C[q_*]) H_C^{\{1/2\}} > 0 \quad (51)$$

$$S_{\text{LPC}}^{\{(2)\}}[q] = \mathcal{K}_0 \int d^4x \sqrt{-g} [-\frac{1}{2} g^{\{\mu\nu\}} \langle \nabla_\mu \wedge C q, Z_C \nabla_\nu \wedge C q \rangle - V_C^{\{(2)\}}(q)] \quad (52)$$

$$V_C(q) = \frac{1}{2} \langle D_C(q), C_N(Y_C(q)) D_C(q) \rangle \quad (53)$$

The local response Hessian follows from derivatives of the same resolvent, rather than from an independent sector response.

$$\partial_A C = -C Y_{,A} C \quad (54)$$

$$\partial_A \partial_B C = C Y_{,A} C Y_{,B} C + C Y_{,B} C Y_{,A} C - C Y_{,AB} C \quad (55)$$

In the minimal single-metric, two-derivative branch, the principal symbol of every nondegenerate fundamental massless block is proportional to one metric null form. Under additional standard assumptions, the geometric gauge block is related to Fierz–Pauli massless spin-2 and the Einstein infrared branch [13–17]. These are conditional physical consequences, not additional BFG fundamental laws.

$$\sigma_2(\Gamma_a)(x, k) = Z_a g^{\{\mu\nu\}} k_\mu k_\nu P_a \quad (56)$$

$$g^{\{\mu\nu\}} k_\mu k_\nu = 0 \quad (57)$$

7.2 Physical realization map

A natural system is associated with the local physical branch only after the universal invariant class has been generated. The empirical realization map tests whether a smooth carrier, nondegenerate soldering, positive closure stiffness, reversible local transport, stable characteristic structure, and a carrier order readout compatible with r_U or ω_U correspond to that class under the Local Physical Closure Principle. These carrier conditions do not alter $\mathbb{U}$.

Physical-stage	status
The existence of an admissible local geometric branch is conditional. The identification of the resulting metric and spectrum with observed spacetime and gravity remains empirical. The standard field-theory literature constrains the carrier interpretation [13–17]; it does not independently derive BFG.	

8. Matter: nested nuclear, atomic, and molecular closure

Atomic and molecular systems are treated as realization tests of already-generated invariant classes, not as definitions of those classes. Established quantum mechanics, electrodynamics, nuclear physics, and chemistry supply the carrier observables [18]. BFG contributes the universal retained/complement, persistence, nesting, and export coordinates against which those observables are compared.

8.1 Atomic realization test

An atomic realization is supported only when an independently measured bound system maps to a universal class with an isolated persistent spectral block, noncollapsed distinguishability, bounded recursive transport, admissible complement/export, and an internally generated dual-order state compatible with the carrier data. The class exists mathematically without the word “atom”; the label is attached only after the empirical map succeeds. No atomic Hamiltonian, threshold, or order parameter is inserted into $\mathbb{U}$.

$$\Pi_{\text{atom}}: N_{\text{atom}} \rightarrow Q_{\mathbb{U}} \text{ is accepted only from independently measured carrier data} \quad (58)$$

This is intentionally a structural implication rather than a derivation of the hydrogen spectrum from a scalar toy load. The no-retuning claim is that the BFG closure law is unchanged; standard physics supplies the carrier Hamiltonian and measured spectral data.

8.2 Molecular and chemical reclosure

A molecular realization is tested by whether independently identified lower bound systems map jointly into a higher universal class whose retained response is nonfactorizable while the lower subclosures remain recoverable. The higher class is generated by $\mathbb{U}$ before chemical naming; chemistry supplies the carrier Hamiltonian and measured relational spectrum.

$$C_{\text{mol}} \neq C_{\text{atom},1} \oplus C_{\text{atom},2} \quad \text{when the higher retained response is nonfactorizable} \quad (59)$$

The distinction between constitutive nesting and temporal generation is important. The hierarchy expresses closure precedence. It does not imply that every molecule must literally replay Level-0 or atomic formation as a temporal developmental sequence.

9. From chemistry to abiogenesis

The molecular-to-biological transition is treated as a falsifiable generativity problem rather than an adjacent pair of labels [5]. External origin-of-life research shows that several carrier-level ingredients are experimentally tractable: activated pyrimidine ribonucleotides can arise under prebiotically plausible chemistry [19], common precursor chemistry can feed RNA, amino-acid, and lipid building blocks [20], template-directed polymer synthesis can occur in model protocells [21,22], protocell populations can exhibit competition [23], and recent autocatalytic systems can couple reaction dynamics to compartment growth and reproduction-like behavior [24]. These results constrain carrier plausibility; none validates BFG.

9.1 Prebiotic realization test

Reaction-network organization, compartment persistence, throughput, renewal, heredity, and repeated-cycle persistence are now treated strictly as empirical coordinates used to map prebiotic systems into the universal quotient. They are not terms in the generator and do not define a prebiotic mathematical class in advance. The only mathematical generator remains Equation (42).

$$\Pi_{\text{prebio}}: N_{\text{prebio}} \rightarrow Q_{\mathbb{U}} \quad (60)$$

$$\varepsilon_{\text{cycle}}(x) = d_{\mathbb{U}}(\Pi_{\text{prebio}}(T_{\text{cycle}} x), [U(X_x)]_{\{\equiv_{\mathbb{U}}\}}) \quad (61)$$

For empirical work, the multicomponent observation vector $q_B, q_F, q_R, q_H, q_C, q_S$ remains useful [5], but its role is diagnostic. Those observables are measured independently and compared with invariant coordinates of $\mathbb{U}$.

generated classes. No component is permitted to change the universal capacity transport, neutral response, positive-metric witness rule, endogenous dual-order mapping, persistence rule, or gate.

$$\vec{q}_{abi} = (q_B, q_F, q_R, q_H, q_C, q_S) \quad (62)$$

A strong empirical test compares the full frozen BFG readout against capacity-matched hierarchy-only, compartment-only, heredity-only, and shuffled-coupling nulls. The repeated-cycle requirement remains: a one-off vesicle or isolated copying event is not enough to identify biological closure [5,25].

10. Cellular, multicellular, and organismic reclosure

10.1 Cellular closure

The previously described biological ordering is retained only as a natural-history and systems-biology comparison sequence [26]. In the Strong Universal mathematics there is no cellular, tissue, organ, or organism generator. There is only the orbit of $\mathbb{U}$ in the target-independent quotient; biological labels are empirical realization assignments made after the invariant classes exist.

A cellular realization therefore requires prospectively measured boundary persistence, throughput, informational renewal, and recursive self-maintenance across turnover to co-localize in one generated invariant class. Failure of that mapping is a failure of the proposed biological realization, not a reason to modify the universal operator.

10.2 Multicellular projection and morphogenesis

A prior organismic formulation treats local cellular closures as projecting into a form/field domain, being read out, differentiated, and reintegrated into a global closure [27]. Classical developmental biology provides independent mechanisms—reaction–diffusion, positional information, morphogen interpretation, gene-regulatory organization, mechanics, and bioelectric pattern control [28–32]. BFG does not replace these mechanisms. The unification claim is that they can be evaluated as carrier realizations of a common closure transition.

$$\Pi_{org}: N_{org} \rightarrow Q_U \quad (63)$$

A multicellular mass does not qualify merely because cell number increases. The organismic predicate requires persistent global integration, preserved local differentiation, admissible projection/readout, regulated export, and failure under the appropriate ablations. The distinction cell growth $\neq$ organism formation is retained [27].

10.3 Organismic closure

Organismic realization is treated as a many-subclosure empirical mapping problem. Local biological closures must remain individually recoverable while a higher nonfactorizable invariant class captures global persistence, differentiation, return, and regulated export. The organism label is assigned only after this higher class has been generated and matched; it never enters the formation rule.

11. Sensorimotor, cognitive, and memory closure

Sensorimotor, learning, memory, self-model, perspective, and identity concepts are likewise downstream realization hypotheses [26]. The universal mathematics instead tracks target-independent quantities such as externally driven distinguishability, cross-channel nonfactorization, recursive return, witness persistence, and self-referential reachability. Only after these coordinates are estimated independently may cognitive labels be attached.

Table 2. Target-independent cognitive invariant coordinates used only for downstream realization tests.

Invariant coordinate	Target-independent meaning
External distinguishability	Externally driven differences remain recoverable after neutral resolution and recursive transport.

Invariant coordinate	Target-independent meaning
Nonfactorization	Cross-channel closure cannot be reduced to a direct sum of independent retained blocks.
Return depth	Perturbations re-enter the same closure through a bounded sequence of return mappings.
Witness persistence	A prospectively selected invariant subspace remains power bounded across separated recursions.
Adaptive plasticity	Transverse degrees of freedom change while a declared witness remains recoverable.
Self-reference reachability	A closure representation can reach and be reached from a representation of its own persistent state.
World/self separation	Two distinguishable relational sectors remain jointly mediated without collapse.
Integration margin	Shared-neutral off-diagonal coupling remains nonzero under preserved differentiation.
Differentiation margin	The active retained operator keeps a positive minimum singular-value margin.

12. Thermodynamic maintenance as a readout, not a second consciousness law

A prior thermodynamic model introduced explicit entropy, information, and energy-balance equations [35]. In the present unification those equations are not used as a second fundamental definition of consciousness. Instead, thermodynamic quantities are read from a physical carrier that realizes the universal retained/complement partition.

For a normalized carrier state D , define retained and complementary quadratic fractions:

$$r_N(D) = \langle D, C_N(Y)D \rangle / \langle D, D \rangle \quad (64)$$

$$b_N(D) = \langle D, B_N(Y)D \rangle / \langle D, D \rangle = 1 - r_N(D) \quad (65)$$

A physical model may then map these fractions into bound energy, exported heat, chemical work, or information-processing costs. Landauer's result constrains the energetic cost of logically irreversible information operations [36], but it does not imply that consciousness is simply entropy reduction.

$$E_bind = \mathcal{R}_E^{\{bind\}}[r_N, carrier] \quad (66)$$

$$E_exp = \mathcal{R}_E^{\{out\}}[b_N, carrier] \quad (67)$$

The readout maps $\mathcal{R}_E$ are empirical constitutive maps, not new fundamental BFG dynamics. They do not enter $\mathbb{U}$ as external energetic controls. The universal map feeds back only the structural state that it generates and its endogenous dual-order readout. A measured carrier may change the represented X and therefore Y through the already declared upstream map, but no domain-specific thermodynamic coefficient is allowed to retune the EDO-URE.

13. Conscious subjectivity as a shared-neutral EDO-URE realization class

The shared-neutral cognitive formulation supplies a positive-load/resolvent family compatible with the universal architecture [12]. In the Endogenous Dual-Order Strong Universal Form, however, “conscious subjectivity” is not a

mathematical stage predicate and the local scalar φ used in the carrier model is not a fundamental universal input. It is an empirical realization coordinate to be compared with a declared readout of r_U or ω_U . A measured cognitive system must map into an invariant class exhibiting simultaneous shared-neutral integration, preserved differentiation, bounded recursion, self-indexing, and identity persistence under fixed rules.

$$Y_\varphi = M_\varphi^\dagger M_\varphi \geq 0 \quad (68)$$

$$C_\varphi = (I + Y_\varphi)^{-1} \quad (69)$$

$$Q_\varphi = Y_\varphi (I + Y_\varphi)^{-1} = I - C_\varphi \quad (70)$$

Relative to a fixed channel decomposition with projectors P_i , exact channel factorization is equivalent for Y_φ and C_φ . Shared-neutral integration therefore requires nonzero off-diagonal response, while differentiation is protected by the smallest singular value.

$$Y_{\{ij\}}(\varphi) = P_i Y_\varphi P_j, \quad C_{\{ij\}}(\varphi) = P_i C_\varphi P_j \quad (71)$$

$$\Xi_\varphi = [\Sigma_{\{i \neq j\}} \|P_i C_\varphi P_j\|^2_{HS}] / \|C_\varphi\|^2_{HS} \quad (72)$$

$$\Delta_\varphi = \sigma_{\min}(C_\varphi|_{\{H_{\text{act}}\}}) = 1/[1 + \lambda_{\max}(Y_\varphi|_{\{H_{\text{act}}\}})] \quad (73)$$

World/self relation, self-indexing, identity persistence, and bounded recursion therefore remain jointly testable coordinates, not definitional labels. The empirical identification of a matching invariant class with conscious subjectivity remains a hypothesis [12].

$$K_\varphi = R_\varphi C_\varphi \quad (74)$$

$$K_\varphi^\dagger L K_\varphi \leq L \quad (75)$$

$$\sup_n \|K_\varphi^n x\|_L < \infty \quad (76)$$

Perturbational-complexity and near-criticality research provide external benchmarks for integration-plus-differentiation and bounded dynamical regimes [42,43]. BFG does not identify its variables directly with PCI or with a generic criticality index; such mappings must be preregistered.

14. Phenomenal content and qualitative identity

Qualitative content is handled in the same order: first the universal equivalence class, then the phenomenological realization claim [44]. The formal object $Q[X]$ is a closure-equivalence class determined by invariant structure. Whether one such class corresponds to experienced redness, pain, or another quale is not built into the quotient and must be established independently.

14.1 BFG closure equivalence

Let $\mathcal{G}_{\text{adm}}$ be the set of admissible transformations that preserve the declared witness spectrum, the neutral-response spectra, the local order state within tolerance, and the restricted recursive transport. Define:

$$X \sim_{\text{BFG}} X' \Leftrightarrow \exists G \in \mathcal{G}_{\text{adm}}: X' = GX \text{ and } \mathbb{I}_U(X') = \mathbb{I}_U(X) \quad (77)$$

$$\mathbb{I}_U(X) := \{\text{Spec } C_N(Y_X), \text{Spec } B_N(Y_X), \omega_U(X), r_U(X), \mathcal{W}_X, R_C|_{\{\mathcal{W}_X\}}\} \quad (78)$$

$$Q[X] := [X]_{\sim_{\text{BFG}}} \quad (79)$$

The formal qualitative identity is therefore an equivalence class of the closure geometry, not an extra postulated substance. For two content closures:

$$Q_{\text{red}} \neq Q_{\text{blue}} \Leftrightarrow [C_{\text{red}}]_{\sim_{\text{BFG}}} \neq [C_{\text{blue}}]_{\sim_{\text{BFG}}} \quad (80)$$

This construction closes the formal operator gap only at the mathematical level. The empirical claim that a particular equivalence class is phenomenal redness still requires content-specific neural and behavioral tests. If closure invariants

fail to distinguish red from blue after controlling local activation, or if the equivalence relation can only be defined using the target report itself, the BFG qualia claim is weakened or fails [44].

14.2 Ownership and selfhood

Ownership is likewise formulated as a relation between a content-equivalence class and a persistent identity witness after both have been generated within the universal quotient. The phrase “my experience” is therefore an empirical interpretation of a bidirectional closure relation, not a primitive term in $\mathbb{U}$.

$$\text{SelfOwned}(Q_i)=1 \Leftrightarrow Q_i \text{ is bidirectionally reachable from the persistent identity witness } \mathcal{W}_{id} \quad (81)$$

External work on bodily self-consciousness supports the broader empirical fact that ownership, self-location, perspective, and multisensory self-processing can dissociate [45]. It does not validate the BFG selfhood operator; it supplies independent phenomena against which the closure architecture can be tested.

15. The complete cosmic precedence map

The cosmic precedence map is retained only as an empirical comparison scaffold. It does not define the order in which $\mathbb{U}$ must produce named natural categories. The mathematical orbit contains unlabeled invariant classes; physical, chemical, biological, cognitive, and phenomenal names are assigned afterward if independent carrier data realize them. Structural precedence therefore remains conceptually useful without becoming a hidden stage predicate.

Table 3. Natural-system realization scaffold mapped onto target-independent EDO-URE invariant coordinates.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
0	Level-0 admissibility	Difference/coherence/neutral capacities and an isolated negative formation mode.
1	Primitive formed structure	Observed carrier candidate maps to a class with a realized Level-0 branch and finite positive load.
2	Neutral-resolved structure	Observed response is consistent with the exact C_N/B_N partition.
3	Persistent structure	Measured dynamics support a prospectively isolated power-bounded witness block.
4	Local physical realization	Independent carrier data satisfy the physical realization conditions associated with a generated class.
5	Bound physical realization	Measured bound structure maps to a spectrally isolated retained class with admissible export.
6	Nuclear/atomic realization	Quantum carrier data map to a nested bound class with persistent spectral separation.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
7	Molecular realization	Chemical data map multiple lower bound realizations into a higher nonfactorizable class.
8	Macromolecular / sequence realization	Measured relational ordering persists across reclosure-relevant perturbations.
9	Informational realization	Alternative sequence states remain discriminable and transportable under fixed carrier mapping.
10	Replication realization	A closure-relevant witness is reproduced into an additional same-level realization.
11	Error-correction realization	Perturbed carrier states return to the same target-independent invariant class.
12	Metabolic throughput realization	Persistent retention coexists with regulated measured export and renewal.
13	Boundary realization	Measured inside/outside partition remains stable while selective exchange persists.
14	Protocellular realization	Boundary, throughput, renewal, heredity, coupling, and repeated-cycle persistence co-map to one class.
15	Cellular realization	A protocellular carrier realization remains recursively self-maintaining across material turnover.
16	Projection realization	Measured persistent complementary structure becomes accessible to another local realization.
17	Readout realization	Projected structure measurably changes another closure-mediated carrier state.
18	Feedback realization	Projection and return form a bounded empirically observed recursive loop.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
19	Network / field realization	Multiple local realizations map jointly into a nonfactorizable higher class.
20	Attractor realization	Perturbed carrier states return to one invariant basin under fixed measurement rules.
21	Differentiation realization	Distinct local basins persist while co-mapping to one higher class.
22	Tissue realization	Many cellular realizations support a persistent spatial-functional higher mapping.
23	Organ realization	Specialized tissue realizations map into a higher persistent functional class.
24	Organ-system realization	Multiple organ realizations jointly map into an integrated higher class.
25	Organism realization	Global empirical organization persists despite internal turnover and differentiation.
26	Sensory realization	External perturbations map to discriminable internal retained coordinates.
27	Integration realization	Multiple channels map to a shared nonfactorizable neutral-load class.
28	Valence realization	Alternatives exhibit reproducibly distinct persistence/export gradients.
29	Action realization	Measured export changes the exterior and returns as structured input.
30	Learning realization	Future measured transition structure changes after prior perturbation/feedback.
31	Memory realization	A witness remains recoverable across temporally separated cycles.
32	Self-model realization	Carrier data support a persistent internal representation of the system's own state.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
33	Perspective realization	World-related and self-related coordinates remain distinct but jointly mediated.
34	Identity realization	A persistent self-index remains while transverse carrier degrees adapt.
35	Conscious-subjectivity realization hypothesis	Independent cognitive data map jointly to integration, differentiation, self/world structure, identity, and bounded recursion.
36	Phenomenal-unity realization hypothesis	Multiple content-related realizations occupy one globally connected differentiated class.
37	Qualitative-content realization hypothesis	Content identity is tested against a target-independent closure-equivalence class.
38	Self-owned experience realization hypothesis	Content-related class is bidirectionally linked to an independently estimated identity witness.
39	Narrative / intersubjective realization hypothesis	Temporally extended self-model and other-subject relations are tested only after class generation.
40	Knowledge / science / civilization realization hypothesis	Collective carrier systems are mapped post hoc into higher invariant classes; highly speculative empirical extension.

16. Executed numerical demonstrations

All simulations in this section were executed for the present whitepaper in Python 3.13.5 using NumPy 2.3.5 and Matplotlib 3.10.8. They are deterministic and dimensionless. No empirical dataset was fitted. The simulations are designed to test algebraic or stability consequences of the declared equations and to expose failure surfaces; they do not demonstrate that any physical, biological, or conscious system realizes BFG.

16.1 Exact neutral-partition surface

For two independent eigenloads $y_1, y_2 \geq 0$, define $c_i = (1 + y_i)^{-1}$ and $b_i = y_i / (1 + y_i)$. The exact partition $c_i + b_i = 1$ holds independently on each eigenmode. A symmetric diagnostic of simultaneous retained/complement availability is:

$$\Gamma_{rc}(y_1, y_2) = \frac{1}{2}[4c_1b_1 + 4c_2b_2], \quad 0 \leq \Gamma_{rc} \leq 1 \quad (82)$$

The numerical sweep used $y_1, y_2 \in [10^{-2}, 10^2]$. The maximum grid value was 0.999790 at $y_1 \approx 0.9715$, $y_2 \approx 0.9715$; the maximum absolute partition error was $2.220e-16$, consistent with floating-point roundoff. The analytic maximum $\Gamma_{rc}=1$ occurs at $y_1=y_2=1$, where retained and complementary fractions are balanced mode by mode.

Figure 2. Executed 3D neutral-partition simulation. The surface is derived solely from the exact BFG resolvent pair. The ridge collapses to the analytic balanced maximum at $y_1=y_2=1$; no sector-specific response function is used.

16.2 Spectral generativity normal form

The spectral gate depends only on the sign of the next-level minimum formation eigenvalue. To visualize this sign change without assigning a physical sector, a two-mode symmetric normal form was used:

$$K_{nf}(\delta, g) = [[1+\delta, -g], [-g, 1-\delta]] \quad (83)$$

$$\lambda_{\min}(K_{nf}) = 1 - \sqrt{(\delta^2 + g^2)} \quad (84)$$

$$g_U = 1 \Leftrightarrow \delta^2 + g^2 > 1 \quad (85)$$

A dense sweep over $\delta \in [-1.6, 1.6]$ and $g \in [0, 1.6]$ reproduced the analytic circular sign boundary exactly to grid resolution. 69.5% of the displayed rectangular parameter window lies in the negative-eigenvalue region; this percentage has no physical meaning and depends only on the chosen plotting window. The scientific content is the visible gate surface $\lambda_{\min}=0$.

Figure 3. Executed 3D spectral-gate normal form. The surface illustrates the canonical rule $g_U=1$ when the candidate next-level formation operator acquires a negative minimum eigenvalue. The matrix is a pedagogical normal form, not a carrier law.

16.3 Order–feedback stability corridor

The declared order-indexed toy Jacobian uses $v(\varphi) = \exp[-4(\varphi-1)^2]$, $\alpha=0.35$, $\beta=0.25$, $d=0.45$, and $c=0.08$ [5]. The present work re-executed the full (φ, B) spectral-radius sweep.

$$v(\varphi) = \exp[-4(\varphi-1)^2] \quad (86)$$

$$J(\varphi, B) = [[B+0.35(1-v), 0.08v], [0.08v, 0.45+0.25(1-v)]] \quad (87)$$

At $B=0.82$ the executed values are $\rho(J) = 0.8366$ at $\varphi=1.0$ and 0.9897 at $\varphi=1.4$. At $B=0.90$ and $\varphi=1.4$, $\rho(J) = 1.0690$, crossing the linear stability boundary. Across the displayed $\varphi \in [0.45, 1.55]$, $B \in [0.55, 0.98]$ grid, 78.9% of sampled points had $\rho < 1$. That fraction is a plotting statistic, not a universal prediction.

Figure 4. Executed 3D spectral-radius surface for the order-indexed interface toy model. The surface reproduces the finite joint stability corridor associated with the cited parameterization [5]. Linear spectral stability is necessary only for this fixed linear toy; nonnormal and time-varying systems require stronger power-bounded or Lyapunov analysis.

16.4 Shared-neutral cognitive integration

The five-channel cognitive simulation was re-executed from the previously defined matrices [12]. The mediation map uses $D = \text{diag}(0.55, 0.65, 0.75, 0.85, 0.95)$, a symmetric five-node ring with three weak chords normalized to spectral norm one, and $M_\varphi = D + g\Phi W$. With $\Phi_{src}=1$ and $v(\varphi) = 1 + 0.45\varphi$, the positive order fixed point is:

$$\varphi^* = [-1 + \sqrt{1 + 4a\Phi_{src}}] / (2a) \quad (88)$$

The executed value is $\varphi^* = 0.7481$. At $g = 0, 0.4, 0.8, 1.2, 1.6$ the integration/differentiation pairs are respectively 0.0: (0.0000, 0.5256); 0.4: (0.0261, 0.4482); 0.8: (0.0819, 0.3442); 1.2: (0.1350, 0.2642); 1.6: (0.1697, 0.2061). These reproduce the previously defined trade-off [12]: coupling raises nonfactorizable integration while lowering the guaranteed differentiation margin.

Figure 5. Executed 3D shared-neutral integration surface Ξ_{φ} . Integration vanishes in the $g=0$ ablation and grows as the common mediation map develops off-diagonal channel coupling. The plot is an internal mathematical demonstrator, not a consciousness measurement.

Figure 6. Executed 3D differentiation surface $\Delta_{\varphi}=\sigma_{\min}(C_{\varphi})$ over the same parameter plane. Increasing shared load can erode the minimum distinguishability margin; this is why BFG does not identify maximal coupling with consciousness.

16.5 Recursive critical surface and derivative identity

At the benchmark $\varphi^*=0.7481$ and $g=0.8$, the previous parameterization [12] uses $R_0=1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$. The present execution gives:

$$K_{\varphi}(r,g)=rR_0C_{\varphi}(g) \quad (89)$$

$\rho(R_0C_{\varphi})=0.8711$, so $r_{\text{crit}}=1/\rho=1.1480$. Choosing $r=1.1422$ yields $\rho(K_{\varphi})=0.995000$. The analytic resolvent derivative $\partial_{\varphi}C_{\varphi}=-C_{\varphi}Y'_{\varphi}C_{\varphi}$ was checked against a centered $h=10^{-6}$ finite difference with relative Frobenius error $4.297e-10$.

Figure 7. Executed 3D recursive stability surface for $K_{\varphi}=rR_0C_{\varphi}$ at fixed $g=0.8$. The critical level $\rho=1$ separates spectrally subcritical and supercritical regions for this fixed linear model.

16.6 Bounded near-critical trajectory

Using the near-critical gain above and the deterministic initial state $s_0=(0.90,-0.55,0.35,0.25,-0.20)$, 160 iterations remained bounded. The initial Euclidean norm was 1.1565, the maximum norm over the run was 1.3535, and the final norm was 0.6575. The trajectory is shown in the first three principal coordinates of the 161 simulated states.

Figure 8. Executed bounded near-critical trajectory projected into three principal coordinates. The run demonstrates power-bounded recurrent behavior for the chosen fixed linear model. It does not demonstrate phenomenal consciousness.

16.7 Endogenous dual-order cross-feeding surface

The reciprocal order law in Equation (33) was evaluated directly over four decades of positive keep and upward structural load. Because the two components are cross-fed and normalized, the surface is scale invariant: only the ratio $\Lambda^{\uparrow}/\Lambda^{\text{keep}}$ determines the order coordinate. At equal loads, $\omega_U^{\text{keep}}=\omega_U^{\uparrow}=0.5$; dominance of one branch drives the order weight of the opposite branch toward unity without creating an additional free parameter.

Figure 9. Executed 3D endogenous dual-order surface. The vertical coordinate is $\omega_U^{\text{keep}}=\Lambda^{\uparrow}/(\Lambda^{\text{keep}}+\Lambda^{\uparrow})$; $\omega_U^{\uparrow}$ is its exact complement. The surface is a mathematical visualization of reciprocal order generation, not a physical-energy measurement.

16.8 Dual-order non-expansion stress test

A deterministic two-mode normal form and a seeded random complex-matrix stress test were used to check the analytic bound in Equation (36). For the two-mode surface, $Y=\text{diag}(y_1,y_2)$, $D=(1,1)/\sqrt{2}$, and the full space is used as the persistent subspace. The displayed grid spans $y_1,y_2 \in [10^{-2},10^2]$. Across that grid the maximum η_{DO} was 0.5765954, below the universal analytic bound $1/\sqrt{2} \approx 0.7071068$.

$$Y=\text{diag}(y_1,y_2), \quad D=(1,1)^T/\sqrt{2}, \quad P_{\mathcal{W}}^{\wedge}(G)=I \quad (90)$$

$$\begin{aligned} \eta_{\text{DO}}(y_1,y_2) &:= \|\tilde{D}^{\wedge}\{\text{DO}\}\|_{\{G \oplus G\}} / \|D\|_G \\ &= [2\Lambda^{\text{keep}}\Lambda^{\uparrow} / ((\Lambda^{\text{keep}}+\Lambda^{\uparrow})\Lambda_0)]^{1/2}, \quad \Lambda_0 := \|D\|_G^2 \end{aligned} \quad (91)$$

Figure 10. Executed 3D two-mode EDO coupling-gain surface η_{DO} . The displayed maximum is 0.5765954; the exact theorem gives $\eta_{\text{DO}} \leq 1/\sqrt{2}$ for every admitted state before the next closure rebuild.

The seeded stress test used 10,000 complex instances with dimensions $d=2,\dots,8$, random positive loads $Y=A^\dagger A/d$, random complex states, and random persistent subspaces projected with $P_{\mathcal{W}^\wedge(G)}$. The maximum observed η_{DO} was 0.6412000821; no instance violated $1/\sqrt{2}$, and the largest numerical error in $\omega_U^{\text{keep}} + \omega_U^\dagger = 1$ was 2.22×10^{-16} . These computations are sanity checks of the stated finite-dimensional construction. They do not prove natural-system realization and they do not identify Λ with physical energy.

16.9 Reciprocal-balance canonicity test

To test whether the reciprocal rule merely looks convenient, a one-parameter power family was compared with the exact balance condition. Let the keep weight be proportional to $(\Lambda^\dagger)^\alpha$ and the upward weight to $(\Lambda^{\text{keep}})^\alpha$. The normalized balance residual is zero for all positive load ratios only at $\alpha=1$, which is precisely the reciprocal rule derived analytically from Equation (32). This numerical surface is illustrative; uniqueness itself follows algebraically from the two balance equations.

$$\omega_{\text{keep}}^\alpha := (\Lambda^\dagger)^\alpha / [(\Lambda^{\text{keep}})^\alpha + (\Lambda^\dagger)^\alpha], \quad (92)$$

$$\omega_\uparrow^\alpha := (\Lambda^{\text{keep}})^\alpha / [(\Lambda^{\text{keep}})^\alpha + (\Lambda^\dagger)^\alpha]$$

$$\mathcal{R}_{\text{bal}}(\alpha) := |\omega_{\text{keep}}^\alpha \Lambda^{\text{keep}} - \omega_\uparrow^\alpha \Lambda^\dagger| / [|\omega_{\text{keep}}^\alpha \Lambda^{\text{keep}} + \omega_\uparrow^\alpha \Lambda^\dagger|] \quad (93)$$

Figure 11. Reciprocal-balance canonicity surface. Across eight decades of the load ratio, the normalized balance residual vanishes identically along $\alpha=1$. Other power exponents fail away from equal loads. The plot illustrates, rather than proves, the algebraic uniqueness of the reciprocal solution.

16.10 Canonical polar-transport stress test

The polar transport rule was tested on 10,000 seeded complex instances with dimensions 2–8. Each instance used a random positive master load, a random G -orthogonal persistent subspace, the exact neutral pair, reciprocal order weights, and the cross-fed analysis operator. The polar partial isometry satisfied $\|J^\dagger J - P_{\text{suppl}}\|_2 \leq 3.405 \times 10^{-15}$. Compression of an independently generated positive capacity remained positive in every case; the smallest observed compressed eigenvalue was 0.00615466. The largest Hermiticity defect was 1.019×10^{-15} . Reciprocal balance residuals stayed below 5.551×10^{-17} , the simplex-sum error below 2.220×10^{-16} , and the largest observed dual-coupler gain was $0.604756719 < 1/\sqrt{2}$. These are numerical checks of exact identities, not empirical evidence for a natural carrier.

Figure 12. Projective endogenous-order surface. The canonical keep weight depends only on the ratio $A^\dagger/\Lambda^{\text{keep}}$, demonstrating scale invariance and the one-degree-of-freedom character of the dual pair. Equal loads give $\omega_{\text{keep}} = \omega_\uparrow = 1/2$ and projective scalar $r_U = 1$.

17. Interpretation of the simulations

The simulations support internal consequences of the Endogenous Dual-Order Strong Universal architecture only. They verify the exact neutral partition numerically, display a target-independent spectral formation boundary, reproduce the declared carrier order-feedback corridor, test shared-neutral integration and differentiation, check the resolvent derivative identity, exhibit bounded near-critical recursion, and directly stress-test the new reciprocal dual-order coupler.

The simulations do not prove natural universality. Their role is to establish calculational coherence and expose failure surfaces for the universal operator itself. The stronger empirical test is whether independent natural-system measurements map into the same invariant quotient and whether observed transitions commute with $\mathbb{U}$ within prospectively fixed error bounds.

18. Falsification architecture

A universal mathematical theory becomes vacuous if a failed realization can always be repaired by modifying the generator, externally adjusting order, or redefining the target class after inspection. The Endogenous Dual-Order Strong

Universal Form therefore separates failures of the mathematical self-map, failures of carrier realization, and failures of empirical interpretation.

Table 4. Primary falsification surfaces of the EDO-URE program.

Failure surface	Decision condition
Formation failure	$\lambda_0 \geq 0$ or the Level-0 negative mode is not isolated.
Positive-geometry failure	A declared Gram-type construction or positivity domain fails, or $Y_{\{n+1\}}$ cannot be reconstructed as positive without changing the universal rule.
Neutral-resolution failure	Empirical response requires an independent pole or nonlinear law not derivable from the universal C_N/B_N family.
Persistence failure	The bounded non-decaying witness subspace is not closed/isolated on the declared carrier, or the required peripheral gap disappears under preregistered tolerances.
Generativity failure	A higher nonterminal iterate requires a target-specific capacity operator, gate, threshold, or synthesis map rather than the canonical polar transport and common formation rule.
Carrier-transfer failure	The empirical realization map must alter the universal generator or invariant quotient after entering a new domain.
Hierarchy-only null	A capacity-matched generic hierarchy predicts the same held-out transitions without BFG mediation/return structure.
Atomic failure	The proposed atomic readout adds no predictive or organizing content beyond relabeling established carrier theory.
Abiogenesis failure	Repeated-cycle closure does not depend on the predicted coupling, export, or witness relations under ablation.
Organism failure	Projection/readout/mediation variables add no prospective value beyond matched developmental nulls.
Consciousness failure	Matched recurrent or connectivity models reproduce held-out perturbational predictions without shared-neutral constraints.
Qualia failure	No non-circular content-specific closure invariant is found, or local carrier activity fully explains content and ownership.
Global no-retuning failure	Any target domain can be fit only by modifying the universal generator, invariant coordinates, quotient metric, or tolerance after seeing the outcome.
Endogenous-order failure	Observed or simulated generator behavior requires violating normalized reciprocal balance, adding an external level-specific order input, or introducing a branch-dependent coefficient not derivable from Equations (32)–(34).

Failure surface	Decision condition
Projection-gain failure	The canonical graph-metric projector or polar partial isometry cannot be defined on the declared active support, or the analytic non-expansion result fails under the stated assumptions.

19. Preregistration and no-retuning protocol

Every empirical realization test must freeze, before target outcomes are inspected, the carrier-to-BFG role extraction, neutral response family, bounded-nondecay witness criterion, peripheral-gap tolerance, positive graph-metric projection, reciprocal-balance readout, canonical polar transport, common formation gate, quotient metric, realization-map class, and all numerical tolerances. None of these objects may be replaced after the target domain or held-out outcome is known.

- Freeze operator family and numerical tolerances before outcome access.
- Separate structural level n from physical/developmental time t .
- Report retention, complement, export, spectral margin, and witness continuity jointly.
- Include hierarchy-only, order-independent, return-ablated, shuffled-coupling, and capacity-matched recurrent controls where applicable.
- Preserve negative outcomes and localize the failed bridge instead of rewriting the stage definition.
- For consciousness and qualia, preregister report-independent or report-minimized carrier estimators and construct the invariant class without using the phenomenal label as a feature.
- For abiogenesis, require repeated-cycle persistence rather than a one-off successful compartment.
- For cross-domain claims, use the same universal quotient coordinates and generator; only the empirical carrier map Π_s may differ, and its form must be declared before target outcomes are inspected. A scalar carrier order coordinate may be used only as a downstream readout of r_U or ω_U , never as a retuned input to $\mathbb{U}$.

20. Relationship to established science

The Endogenous Dual-Order Strong Universal Form is not required to compete with mature theories by verbal similarity. The previous canonical-reduction layer already required target-independent quotients, but still left a formal question: which invariant stratum, congruence, iterate aggregation, representation map, and parameter readout should be used? The Freedom-Closed edition removes those discretionary choices. One intrinsic behavioral quotient is defined before any mature theory is named; exact theory recovery is then an isomorphism test against a pre-existing quotient dynamics, not a quotient-design problem.

The burden is therefore sharper than explanatory compatibility. BFG must generate the branch, behavioral equivalence, quotient dynamics, effective parameter values, and observable algebra without target-specific selection. If several inequivalent parameter values, quotient identifications, or observable realizations survive, no preferred solution is chosen: the claimed reduction remains below closure. Residual nonuniqueness is therefore a falsifiable failure condition, not a tuning opportunity.

20.1 Intrinsic behavioral quotient: removal of stratum and congruence choice

For each state X , let $\Xi_U(X)$ denote the complete canonical datum generated by the universal grammar: positive master load, exact neutral resolvents, recursive transport, intrinsic persistent subspace, positive-metric projection, reciprocal dual-order state, cross-fed analysis operator, polar support transport, reconstructed Gram geometry, and every other generator object fixed by the preceding equations. Two realizations of Ξ_U are identified only when they are related by a simultaneous structure-preserving representation isomorphism. The full future BFG behavior of X is the sequence of these canonical isomorphism classes along every iterate.

$$\Xi_U(X) := \text{complete canonical BFG datum generated} \quad (94)$$

from X by the fixed universal operator grammar

Future-behavioral equivalence is defined globally rather than selected per target theory. Equality of complete future canonical data is preserved by shifting the iterate index, so the relation is automatically a $\mathcal{U}$ -congruence. It is the coarsest identification that removes representation redundancy while preserving every future distinction expressible by the frozen canonical BFG data stream.

$$\mathcal{B}_U(X) := ([\Xi_U(\mathcal{U}^n X)]_{\cong})_{n \in \mathbb{N}_0} \quad (95)$$

$$X \equiv_U X' \Leftrightarrow \mathcal{B}_U(X) = \mathcal{B}_U(X') \quad (96)$$

$$X \equiv_U X' \Rightarrow \mathcal{U}(X) \equiv_U \mathcal{U}(X') \quad (97)$$

$$Q_U := X_{\text{adm}} / \equiv_U, \quad \tilde{\mathcal{U}}_U([X]) := [\mathcal{U}(X)] \quad (98)$$

The quotient Q_U and its induced map $\tilde{\mathcal{U}}_U$ therefore require no chosen stratum S and no chosen equivalence relation $\approx$. For any supplied initial class, the only admissible reduction branch is its own orbit closure under $\tilde{\mathcal{U}}_U$. A theory label cannot alter that branch.

20.2 No-choice exact-law, parameter, and observable recovery

Represent a candidate mature theory by a parameterized dynamical system $T_\lambda = (S_T, F_{\{T, \lambda\}}, \mathcal{O}_T)$. No iterate multiplier m is fitted: recovery is tested against one canonical BFG recursion step and therefore against all of its iterates. A map Ψ is not a tunable BFG object; it is only a witness that the already-generated BFG orbit closure and the target dynamical system are isomorphic (or, where explicitly stated, exactly semiconjugate). If two witnesses differ only by an automorphism commuting with the quotient dynamics, their difference is representation gauge rather than physical freedom.

$$\mathcal{C}_U([X]) := \text{closure}\{ \tilde{\mathcal{U}}_U^n([X]) : n \in \mathbb{N}_0 \} \quad (99)$$

$$T_\lambda := (S_T, F_{\{T, \lambda\}}, \mathcal{O}_T) \quad (100)$$

Target parameters are not supplied by a freely chosen readout map. Exact conjugacy itself defines the set of parameter values compatible with the pre-existing BFG quotient dynamics. Level-4 closure is passed only when this set is a singleton after quotienting by unit conventions. If multiple physically inequivalent values remain, BFG has not derived that parameter and no preferred value may be selected.

$$\Psi \circ \tilde{\mathcal{U}}_U = F_{\{T, \lambda\}} \circ \Psi \quad \text{on } \mathcal{C}_U([X]) \quad (101)$$

$$\Theta_T([X]) := \{ \lambda : \exists \Psi \text{ satisfying (101) and preserving} \quad (102)$$

the canonical observable algebra }

Likewise, no independent observable bridge R_T is admitted. The target observable algebra must pull back under the conjugacy to the canonical observable algebra generated by Ξ_U on the BFG orbit closure. Unit changes and basis changes are representation conventions; they cannot modify spectra, dimensionless ratios, causal relations, closure ranks, or any other invariant prediction.

$$\Theta_T([X]) / \sim_{\text{units}} = \{ [\lambda_U] \} \quad (103)$$

20.3 Freedom-closure ledger and no-tie-break rule

The table below distinguishes fixed axioms and representation conventions from discretionary model choices. Every internal choice that could permit post-hoc rescue is either replaced by a unique construction or converted into a failure condition. The admissible category and initial condition are inputs to the theory statement; basis choice, units, and instrument noise are representation/measurement conventions and are never allowed to modify $\mathbb{U}$.

Table 5. Freedom-closure ledger for the Unified V3 generator and reduction layer.

Potential freedom	Freedom-closed rule	Residual status
External order parameter	Reciprocal balance uniquely determines the dual-order pair from persistent loads.	None in $\mathbb{U}$.
Persistence window / contour	Use the intrinsic bounded non-decaying invariant subspace; contour only represents an already isolated block.	No tunable window.
Projection metric	$G=I+Y$ fixes the unique G -orthogonal persistent projector.	Unique.
Polar-kernel extension	Moore-Penrose support polar factor with zero action on $\ker(\mathcal{A}_{DO})$.	Unique.
Next-level geometry	Rebuilt by the same positive Gram construction.	Unique once X is given.
Stage predicate / target label	Absent from $\mathbb{U}$ and from quotient construction.	Downstream naming only.
Stratum S	Removed: use global Q_U and the orbit closure of the supplied class.	No target-selected stratum.
Congruence $\approx$	Removed: $\equiv_U$ is equality of the complete future canonical data stream.	Unique.
Iterate multiplier m	Removed: exact recovery is one-step; all n -step relations follow by iteration.	No time-step fitting.
Representation Ψ	Only an isomorphism witness; alternatives differing by commuting automorphisms are gauge.	Gauge only.
Target parameters λ	Exact conjugacy defines Θ_T ; recovery requires one physical class modulo units.	Nonunique $\Rightarrow$ L4 fails.
Target observable map	No free R_T ; target observables must pull back to the canonical BFG observable algebra.	Failure $\Rightarrow$ L4 fails.
Mathematical tolerance ε	Theorem conditions are exact; no ε enters the generator or quotient.	Empirical uncertainty only.
Degenerate tie-break	No inequivalent preferred branch/value may be chosen.	Ambiguity remains a failure.

20.4 Five-level natural-recovery certificate

To make the phrase “falls naturally out of BFG” non-rhetorical, the paper adopts five sequential closure obligations. They are not subjective ratings. A later level cannot be claimed unless every earlier level has been passed with a frozen construction.

Table 6. Five-level natural-recovery certificate for theory emergence from $\mathbb{U}$.

Level	Natural-recovery obligation	Pass condition	What failure means
L1	Endogenous branch	The branch is the orbit closure of an already-generated class in the single intrinsic quotient Q_U .	Only analogy; no BFG reduction yet.
L2	Intrinsic quotient	Future-behavioral equivalence $\equiv_U$ and $\bar{U}_U$ are fixed globally before the target theory is named.	A target-specific coarse-graining would be required.
L3	Exact one-step law recovery	The target dynamics is exactly conjugate/semiconjugate to $\bar{U}_U$ on the canonical orbit closure with no fitted iterate multiplier.	The known law is imported or only approximately analogous.
L4	Unique parameter and observable closure	Exact recovery selects one physical parameter class modulo units; target observables pull back to the canonical BFG observable algebra.	A hidden parameter/readout freedom remains.
L5	Independent natural realization	Measured carrier transitions commute with the frozen quotient dynamics within preregistered instrument-derived uncertainty.	The mathematics is not yet a natural realization.

$$C_T := (C_1, C_2, C_3, C_4, C_5) \in \{0, 1\}^5 \quad (104)$$

$$L_{\text{nat}}(T) := \sum_{j=1}^5 \prod_{k=1}^j C_k \quad (105)$$

Equation (105) prevents skipped levels: if any prerequisite fails, all later contributions vanish. A value $L_{\text{nat}}=5$ therefore means that the full chain from target-independent branch formation to independent natural realization has been completed. The universal architecture is designed to make such a five-level reduction possible without changing $\bar{U}$; it does not assign $L_{\text{nat}}=5$ to a sector merely by declaration.

20.5 Finite quotient stabilization theorem

The quotient architecture already has a nontrivial exact consequence that is independent of any named physical or biological theory. If Q is finite, repeated quotient reclosure produces a descending image chain that must stabilize after finitely many steps.

$$Q_0 := Q, \quad Q_{k+1} := \bar{U}(Q_k), \quad Q_0 \supseteq Q_1 \supseteq Q_2 \supseteq \dots \quad (106)$$

$$Q_\infty := \bigcap_{k \geq 0} \text{Im}(\bar{U}^k) = Q_N \text{ for some finite } N \quad (107)$$

$$\bar{U}|_{Q_\infty} : Q_\infty \rightarrow Q_\infty \text{ is bijective} \quad (108)$$

The last statement follows because $\bar{U}$ maps the stabilized finite image onto itself; surjectivity on a finite set implies bijectivity. This gives an exact finite analogue of the BFG transient-to-persistent split: many-to-one transient reduction can feed a recursively persistent sector without requiring an inverse map on the discarded microscopic distinctions.

20.6 Established-theory recovery obligations without importing laws

The new reduction layer does not decree that mature theories have already been recovered. It specifies the proof objects that must exist if they are to emerge naturally. The same certificate is used for every domain; no domain is allowed to replace $\bar{U}$ by its familiar law and then relabel that law as BFG.

Table 7. Domain-independent exact-reduction obligations for representative mature theories and frameworks.

Candidate reduction	Exact object that must be inherited from $\mathbb{U}$	No-fit completion required
Special relativity	A quotient with one common Lorentzian causal structure and Poincaré-covariant local propagation.	Lorentz symmetry and observable interval structure must be quotient invariants, not carrier assumptions.
General relativity	A smooth infrared geometric quotient whose effective spin-2 dynamics is Einstein-equivalent.	Newton coupling, cosmological term, matter coupling, and conservation law must be derived readouts.
Quantum mechanics	A complex spectral quotient with norm-preserving/unitary evolution and a canonical measurement readout.	Schrödinger evolution and probability rule must follow from quotient structure rather than be postulated.
QFT / Standard Model	A local field quotient with gauge representation, causal propagation, and quantum observable algebra.	Gauge group, couplings, masses, and scattering observables must be fixed by BFG invariants.
Thermodynamics	A coarse quotient in which retention/complement/export yields entropy and flux balance.	First/second-law structure and transport coefficients must be readouts, not independent axioms.
Free-energy / active inference	An overdamped probabilistic quotient whose inherited flow is an exact variational gradient flow.	The variational functional, metric/mobility, and generative-state identification must be derived.
Evolution / replication	A population-level quotient of replicating witness classes with inheritance, variation, and differential persistence.	Replicator/selection dynamics must be induced from reclosure, not appended.
String-like dynamics	A rank-one extended persistent quotient with a two-dimensional history and induced worldsheet geometry.	Nambu–Goto/Polyakov-type dynamics and any quantum completion must be consequences of the same quotient.
Conscious subjectivity	A shared-neutral, differentiated, self-indexed bounded recursive quotient with operational invariants.	Phenomenal identification requires independent realization data and cannot define the quotient.

Conditional Theorem — Intrinsic Behavioral Quotient. Fix the admissible BFG category and the universal operator grammar. The complete future canonical data stream $\mathfrak{B}_U$ defines a unique global equivalence $\equiv_U$ by Equation (96), and Equation (97) makes it a $\mathbb{U}$ -congruence. Hence Q_U and $\bar{U}_U$ in Equation (98) are well defined without selecting a target-dependent stratum or congruence. For any supplied initial class, the orbit closure in Equation (99) is fixed by $\bar{U}_U$ alone. A mature theory can therefore enter only by satisfying the one-step exact recovery and uniqueness conditions (101)–(103); no alternate quotient may be chosen to rescue a failed reduction.

Conditional Theorem — Freedom-Closed Canonical Reduction. Fix the admissible BFG category and universal operator grammar. Then Ξ_U , the future-behavioral signature $\mathfrak{B}_U$, the equivalence $\equiv_U$, the quotient Q_U , its induced dynamics $\bar{U}_U$, and the orbit closure $\mathcal{C}_U([X])$ are determined by X without a target-theory choice. For a parameterized target family T_λ , exact one-step conjugacy determines Θ_T . A natural Level-4 reduction exists only when Θ_T contains exactly one physical class modulo unit conventions and the target observable algebra is the pullback of the canonical BFG observable algebra. Multiple inequivalent solutions are not optimized or preferred; they are failures of closure. Thus all internal discretionary reduction freedoms are eliminated or converted into explicit failure surfaces. Representation basis, units, initial condition, and empirical measurement error remain external coordinates/inputs rather than tunable laws.

21. Strong Universal closure certificate and scope separation

Within the declared BFG operator category, the principal mathematical sources of discretionary freedom are now closed explicitly rather than shifted downstream. Positive geometry is generated as $K^\dagger K$ and rebuilt by a Gram construction; persistence is the bounded non-decaying recursive subspace; the Riesz contour is representational rather than selectable; dual order is uniquely fixed by reciprocal balance; next-level synthesis is the support polar partial isometry; terminal cases are absorbed by $\perp$; and the neutral core extends to separable Hilbert-space quadratic-form objects. The effective-theory layer uses the single intrinsic quotient Q_U defined by $\equiv_U$, not a target-chosen family of quotients. Exact recovery is one-step, parameter multiplicity is forbidden except for unit gauge, and no independent observable bridge is admitted.

Table 8. Strong Universal closure certificate and empirical scope separation.

Item	Canonical Closure status
Dual-order origin	Closed mathematically: ω_U is the unique normalized solution of reciprocal load balance, not a fitted two-parameter law.
Persistence selection	Closed mathematically: $\mathcal{W}$ is defined by bounded non-decay; an isolated peripheral Riesz projector represents that subspace and is contour-independent.
Positive master geometry	Closed mathematically: $Y=K^\dagger K \geq 0$ and the next Y is regenerated by the same positive Gram construction.
Capacity transport	Closed mathematically on the active support: J is the polar partial isometry of the cross-fed analysis operator; no free next-level transport matrix is introduced.
Terminal cases	Closed mathematically: loss of dual support, persistence gap, positivity, or the common formation gate maps to the absorbing state $\perp$.
Infinite-dimensional carrier spaces	Closed at the resolvent level for separable Hilbert-space quadratic-form objects with closed densely defined K ; C_N and B_N remain bounded positive contractions.
Stage classification	Closed structurally: named stages occur neither in $\mathbb{U}$ nor in $\equiv_U$; one global future-behavioral quotient is constructed before labels.
Dimensional units	Not part of the generator. Unit-bearing physical quantities are downstream carrier readouts and cannot alter $\mathbb{U}$.
Natural representation	Empirical realization hypothesis: Π_s must be fixed from measured carrier observables before transition outcomes are tested.
Atomic / chemical / biological realization	Empirical realization hypotheses: success requires fixed-rule commutation with $\mathbb{U}$ and failure cannot be repaired by generator retuning.
Consciousness / qualia realization	Empirical realization hypotheses: invariant-class membership does not by itself establish phenomenality; independent neural, behavioral, and report-linked tests are required.

Item	Canonical Closure status
Physical-energy identification	Explicitly not claimed by the structural closure loads Λ . A joule-valued energy readout requires an independently validated carrier map and conservation accounting.
Natural-theory reduction	Closed mathematically at the reduction layer: every admissible effective theory must arise on a target-independent $\mathbb{U}$ -congruence quotient with unique inherited dynamics; exact-law recovery, parameter closure, and empirical realization remain separately testable obligations.
Reduction quotient	Closed: $\mathbb{Q}_U$ is fixed by future-behavioral equivalence of the complete canonical data stream; no target-specific S or $\approx$ remains.
Time aggregation	Closed: fitted iterate multiplier m removed; recovery is one-step and therefore fixes every iterate.
Target parameters	Closed conditionally: exact conjugacy must select one parameter class modulo units; otherwise the reduction fails.
Target observables	Closed conditionally: no free observable bridge; pullback must equal the canonical BFG observable algebra on the recovered branch.
Tie-break freedom	Closed procedurally: multiple inequivalent solutions are never optimized or preferred; they are registered as non-closure.

22. Universal realization and validation program

The realization program follows the freedom-closed five-level certificate. L1 uses only an orbit closure already present in $\mathbb{Q}_U$. L2 requires the intrinsic future-behavioral quotient. L3 requires one-step exact law recovery. L4 requires a unique physical parameter class modulo units and canonical observable pullback. L5 alone introduces measurement: empirical tolerances must come from preregistered instrument/noise models and may not enter $\mathbb{U}$, $\equiv_U$, $\bar{\mathbb{U}}_U$, or the target-parameter solution.

This organization removes a common ambiguity in universal theories: empirical non-realization cannot be repaired by changing the mathematics after the fact, while mathematical closure cannot be presented as empirical confirmation. The universal theorem is therefore closed over its declared category, and the natural-universality claim is a collection of explicit falsifiable realization hypotheses. Scientific vulnerability is retained intentionally at the empirical boundary rather than hidden as an adjustable internal degree of freedom.

23. Compact Canonical Strong Universal Self-Closure Theorem

Conditional	Theorem	—	Canonical	Endogenous	Dual-Order	Strong	Universal	Self-Closure
Assume: (i) admissible Level-0 formation with an isolated determinate branch; (ii) exact neutral elimination with a nonnegative self-adjoint master load; (iii) the canonical bounded non-decay witness definition and an isolated peripheral spectral block; (iv) projection orthogonal in the positive graph metric; (v) reciprocal balance $\omega_{\text{keep}}\Lambda_{\text{keep}}=\omega\uparrow\Lambda\uparrow$ together with simplex normalization; (vi) polar support transport $Q=(ADO\uparrow ADO)^{\wedge\{1/2\}}$, $J=ADOQ^{\wedge+}$; (vii) positive Gram-type reconstruction of the next master load; and (viii) one common formation gate with failure mapped to $\perp$. Then $\mathbb{U}\bar{X}_{\text{adm}}\rightarrow\bar{X}_{\text{adm}}$ is a totalized self-map, the dual-order state is unique whenever both persistent loads are positive, the dual coupler obeys $\ \ddot{D}O\ ^2\leq\ D\ ^2/2$ in the master graph norm, the active-support transport is canonical up to the unique polar partial isometry, and positive master geometry is regenerated at every nonterminal iterate. The same neutral resolvent core remains valid for separable Hilbert-space quadratic-form objects generated by closed densely defined K. No natural-stage predicate, external order scalar, tunable persistence contour, or free next-level synthesis map is part of the generator. Natural realization and physical-energy identification are downstream empirical hypotheses and are not premises of the theorem.								

Conditional Theorem — Compact Canonical Strong Universal Self-Closure. Under the assumptions and exact constructions already stated for Level-0 admissibility, positive master geometry, exact neutral elimination, bounded non-decaying persistence, reciprocal balance, positive-metric projection, support polar transport, Gram reconstruction, and the universal formation gate, the compact operator equation (BFG-U) defines the same totalized self-map $\mathbb{U}$ as the expanded EDO-URE. Its neutral contrast Z_X is uniquely determined by Y_X and is bijectively equivalent to the pair (C_N, B_N) subject to $C_N + B_N = I$. Its split operator S_X is therefore fixed by X , and $\mathfrak{R}_X$ contains only previously canonical operations. Consequently the compact form is fully expandable, freedom-closed, stage-free, and identical in generated orbits to the expanded form. Natural-theory reduction remains governed by the single intrinsic quotient Q_U and the same no-choice recovery conditions; compact notation changes no empirical or epistemic status.

24. Conclusion

Compact Canonical Unified V4 preserves the Freedom-Closed strength of Unified V3 while reducing the canonical generator to a minimal exact operator statement. The neutral pair is encoded losslessly by $Z_X = (I - Y_X)(I + Y_X)^{-1}$, persistence and reciprocal order are combined in S_X , and the downstream polar/Gram/gate sequence is represented by the fully defined reclosure composition $\mathfrak{R}_X$. The fundamental displayed law is therefore $\mathbb{U}[X] = \mathfrak{R}_X(S_X D[X])$, with $X_{n+1} = \mathbb{U}[X_n]$. Every symbol in that equation expands uniquely into the prior canonical construction.

The universal generator therefore contains neither named natural stages nor hidden carrier-specific order, contour, synthesis, or positivity choices. Positive geometry is rebuilt, failure is totalized, and the resolvent core extends beyond finite matrices through nonnegative self-adjoint quadratic forms. Physical, chemical, biological, cognitive, conscious, and phenomenal labels remain downstream hypotheses on target-independent invariant classes. This separation is intentional: it closes the mathematical architecture without converting mathematical closure into an empirical claim.

The resulting universality statement is both shorter and sharper. One positive load produces one bounded neutral contrast; that contrast produces the exact complementary pair; persistence selects what can recurse; reciprocal balance fixes how the persistent pair is cross-fed; canonical reclosure rebuilds the next admissible state or terminates at $\perp$. No domain name, carrier law, target theory, free order parameter, selectable quotient, fitted iterate multiplier, or hidden rescue rule enters the generator. The expanded form remains available for proof and implementation, while (BFG-U) serves as the compact universal BFG equation.

Appendix A. Core formula register

Table 9. Compact formula register.

Role	Formula / statement
Level-0 source	$\mathcal{T}_0=(\mathcal{D}_0, \mathbf{c}_0, \mathbf{N}_0); \mathbf{K}_0=\mathbf{c}_0+\mathbf{N}_0-\mathcal{D}_0$
Formation	$\delta^*_{\pm}=\pm\sqrt{(-\lambda_0)\mathbf{c}_0}$ for simple $\lambda_0<0$
Positive load	$\mathbf{Y}=\mathbf{A}\dagger\mathbf{A}\geq 0; \mathbf{Y}_\text{C}=\mathbf{L}_\text{C}\dagger\mathbf{H}_\text{CL}_\text{C}$
Neutral retention	$\mathbf{C}_\text{N}(\mathbf{Y})=(\mathbf{I}+\mathbf{Y})^{-1}$
Complement	$\mathbf{B}_\text{N}(\mathbf{Y})=\mathbf{Y}(\mathbf{I}+\mathbf{Y})^{-1}=\mathbf{I}-\mathbf{C}_\text{N}(\mathbf{Y})$
Exact partition	$\mathbf{C}_\text{N}+\mathbf{B}_\text{N}=\mathbf{I}$; quadratic retained+complement budget exact
Persistence	$\mathbf{R}_\text{Ce}_\text{T}=\mathbf{e}_\text{T}$; stable complement strictly contractive in a Lyapunov norm
Carrier order readout	$\varphi_\text{obsv}(\varphi_\text{obs})=\Phi_\text{src}$ is a downstream carrier consistency relation; φ_obs must be mapped prospectively from $\mathbf{r}_\text{U}$ or ω_U without entering $\mathbb{U}$.
Persistent metric subspace	$\mathcal{W}$ is the bounded non-decaying recursive subspace; isolated peripheral spectrum gives its unique Riesz representation. $\mathbf{G}=\mathbf{I}+\mathbf{Y}$ defines the orthogonal projector $\mathbf{P}_\mathcal{W}^\wedge(\mathbf{G})$.
Keep / upward / export	$\mathbf{P}_\mathcal{W}^\wedge(\mathbf{G})\mathbf{C}_\text{ND}; \mathbf{P}_\mathcal{W}^\wedge(\mathbf{G})\mathbf{B}_\text{ND}; (\mathbf{I}-\mathbf{P}_\mathcal{W}^\wedge(\mathbf{G}))\mathbf{B}_\text{ND}.$
Endogenous dual order	Reciprocal balance: $\omega_\text{keep}\Lambda_\text{keep}=\omega\uparrow\Lambda\uparrow$ and $\omega_\text{keep}+\omega\uparrow=1$, hence $\omega_\text{U}=(\Lambda\uparrow/\mathbf{S}, \Lambda_\text{keep}/\mathbf{S})$; $\mathbf{r}_\text{U}=\Lambda\uparrow/\Lambda_\text{keep}.$
Cross-fed packet and bound	$\tilde{\mathbf{D}}\mathbf{D}\mathbf{O}=\sqrt{\omega_\text{keep}}\mathbf{D}_\text{keep}\oplus\sqrt{\omega\uparrow}\mathbf{D}_\text{up}; \ \tilde{\mathbf{D}}\mathbf{D}\mathbf{O}\ _\text{G}^2=\{ \mathbf{G}\oplus\mathbf{G} \} =2\Lambda_\text{keep}\Lambda\uparrow/\mathbf{S}\leq\ \mathbf{D}\ _\text{G}^2/2.$
EDO-URE	$\mathcal{A}\mathbf{D}\mathbf{O}$ is fixed by neutral/persistent branches; $\mathbf{J}=\mathcal{A}\mathbf{D}\mathbf{O}(\mathcal{A}\mathbf{D}\mathbf{O}\dagger\mathcal{A}\mathbf{D}\mathbf{O})^\wedge\{\dagger 1/2\}$; $\mathbf{X}_{n+1}=\mathbb{U}[\mathbf{X}_n]$ after canonical capacity compression, positive rebuild, and the common gate.
Iteration	$\mathbb{U}:\tilde{\mathbf{X}}_\text{adm}\rightarrow\tilde{\mathbf{X}}_\text{adm}, \quad \mathbb{U}(\perp)=\perp, \quad \mathbf{X}_n=\mathbb{U}^n[\mathbf{X}_0],$ $\varphi_\text{U}\{\mathbf{U}, n+1\}=\mathcal{O}_\text{U}[\mathbb{U}^n(\mathbf{X}_0)].$
Cognitive load	$\mathbf{Y}_\varphi=\mathbf{M}_\varphi\dagger\mathbf{M}_\varphi$
Cognitive integration	$\Xi_\varphi=\Sigma_{\{i\neq j\}}\ \mathbf{P}_\text{iC}_\varphi\mathbf{P}_\text{j}\ _\text{HS}^2/\ \mathbf{C}_\varphi\ _\text{HS}^2$
Differentiation	$\Delta_\varphi=\sigma_\text{min}(\mathbf{C}_\varphi \mathbf{H}_\text{act})$
Universal invariant quotient	$\mathbf{Q}_\text{U}=\tilde{\mathbf{X}}_\text{adm}/\Xi_\text{U}$; natural labels enter only through external realization maps Π_s .
Canonical data stream	$\mathfrak{B}_\text{U}(\mathbf{X})=(\llbracket \Xi_\text{U}(\mathbb{U}^n\mathbf{X}) \rrbracket_\approx)_{n\geq 0}$

Role	Formula / statement
Intrinsic behavioral equivalence	$X \equiv UX'$ iff $\mathfrak{B}_U(X) = \mathfrak{B}_U(X')$; congruence follows automatically
Canonical quotient	$Q_U = \bar{X}_{\text{adm}} / \equiv_U$; $\bar{U}_U([X]) = [U(X)]$
Quotient commutation	One global quotient induced by $\bar{U}$; no target-specific quotient map
Exact-law recovery	$\Psi \circ \bar{U}_U = F_{\{T, \lambda\}} \circ \Psi$ on canonical orbit closure; no fitted iterate multiplier
Parameter closure	$\Theta_T([X]) = \{\lambda: \text{exact canonical recovery exists}\}$; L4 requires singleton modulo units
Observable closure	Target observable algebra pulls back to canonical BFG observable algebra; no free R_T
Natural reduction	Intrinsic quotient + exact one-step law + unique parameters + canonical observables + independent realization
Five-level certificate	$C_T = (C_1, \dots, C_s)$; $L_{\text{nat}} = \sum_j \prod_{\{k \leq j\}} C_k$
Finite stable image	$Q_\infty = \bigcap_{\{k \geq 0\}} \text{Im}(\bar{U}^k) = Q_N$; $\bar{U} _{Q_\infty}$ bijective
Compact positive load	$Y_X = K[X] \dagger K[X] \geq 0$
Neutral contrast	$Z_X = C_N - B_N = (I - Y_X)(I + Y_X)^{-1}$
Exact pair recovery	$C_N = \frac{1}{2}(I + Z_X)$, $B_N = \frac{1}{2}(I - Z_X)$
Compact split	$S_X = \frac{1}{2} \Omega_X [P_X(I + Z_X); P_X(I - Z_X)]$
Compact universal equation	$\bar{U}[X] = \mathfrak{R}_X(S_X D[X])$; $X_{\{n+1\}} = \bar{U}[X_n]$; $\bar{U}(\perp) = \perp$
Contrast spectrum	$z(y) = (1-y)/(1+y)$, $-1 < z \leq 1$ for $y \geq 0$

Appendix B. Simulation reproducibility register

Table 10. Exact numerical specification for executed figures.

Item	Specification
Software	Python 3.13.5; NumPy 2.3.5; Matplotlib 3.10.8.
Neutral surface	y_1, y_2 logarithmically sampled from 10^{-2} to 10^2 ; 160×160 grid; $\Gamma_{\text{rc}} = \frac{1}{2}(4c_1 b_1 + 4c_2 b_2)$.
Order-feedback surface	$\varphi \in [0.45, 1.55]$, $B \in [0.55, 0.98]$; $\alpha = 0.35$, $\beta = 0.25$, $d = 0.45$, $c = 0.08$; 180×180 grid.
Spectral gate normal form	$\delta \in [-1.6, 1.6]$, $g \in [0, 1.6]$; $\lambda_{\text{min}} = 1 - \sqrt{(\delta^2 + g^2)}$; 180×180 grid.
Cognitive W	Unit-weight ring edges (1,2),(2,3),(3,4),(4,5),(5,1); chords $W_{13} = 0.35$, $W_{24} = 0.25$, $W_{35} = 0.30$; normalized to $\ W\ _2 = 1$.
Cognitive D	$\text{diag}(0.55, 0.65, 0.75, 0.85, 0.95)$.
Order fixed point	$\Phi_{\text{src}} = 1$; $a = 0.45$; $\varphi^* = 0.7481333923$.
Integration surface	$\varphi \in [0.45, 1.25]$, $g \in [0, 1.8]$; 140×140 grid.
Recursive primitive	$R_o = 1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$.
Near-critical gain	$r = 0.995/\rho(R_o C_\varphi) = 1.1422297734$ at $\varphi^* = 0.7481333923$, $g = 0.8$.

Item	Specification
Derivative check	Centered finite difference $h=10^{-6}$; relative Frobenius error 4.2967×10^{-10} .
Trajectory	Initial $s_0=(0.90,-0.55,0.35,0.25,-0.20)$; 160 iterations; PCA obtained by SVD of centered 161-state matrix.
Dual-order surface	$\Lambda_{\text{keep}}, \Lambda_{\uparrow}$ sampled logarithmically from 10^{-2} to 10^2 on a 180×180 grid; $\omega_U^{\text{keep}} = \Lambda_{\uparrow} / (\Lambda_{\text{keep}} + \Lambda_{\uparrow})$, $\omega_U^{\uparrow} = 1 - \omega_U^{\text{keep}}$.
Two-mode EDO gain surface	$Y = \text{diag}(y_1, y_2)$, $y_1, y_2 \in [10^{-2}, 10^2]$ logarithmic 180×180 grid; $D = (1, 1)^{1/2}$; $P_{\mathcal{W}^{\uparrow}(G)} = I$; $\eta_{\text{DO}} = \ \tilde{D}^{\uparrow}\{\text{DO}\}\ _{\{G \oplus G\}} / \ D\ _G$.
Random complex EDO stress test	Seed 20260908; 10,000 instances; dimensions 2–8; $Y = A^{\dagger}A/d$ from complex Gaussian A ; random complex D and random subspace rank $1 \dots d$; G -orthogonal projection. Observed maximum $\eta_{\text{DO}} = 0.6412000821$; 0 violations of the analytic $1/\sqrt{2}$ bound; max simplex-sum error 2.22×10^{-16} .
Reciprocal-balance canonicity surface	$\log_{10}(\Lambda_{\uparrow}/\Lambda_{\text{keep}}) \in [-4, 4]$, $\alpha \in [0.2, 2.0]$; normalized residual $\mathcal{R}_{\text{bal}}$. The analytic zero-residual condition is $\alpha = 1$.
Canonical polar-transport stress test	Seed 20260908; 10,000 complex instances; dimensions 2–8; random positive Y , random G -orthogonal subspaces, SVD polar factor. Max $\ J^{\dagger}J - P_{\text{suppl}}\ _2 = 3.405 \times 10^{-15}$; max balance residual $= 5.551 \times 10^{-17}$; max $\eta_{\text{DO}} = 0.604756719$.

Appendix C. Source-credit register

Table 11. Source-line provenance used by the Unification Paper.

Source line	Use in present paper
Marcel Theodor Wendé & Thomas W. Loker [3]	Co-authored Level-0 admissibility and pre-polar formation source line.
Marcel Theodor Wendé & Luke Kenneth Casson Leighton [4]	Recursive conjugate-closure source line $R_C = \text{MRM}^{\dagger}$, $M = \sqrt{N}$ and related stability geometry.
Canonical BFG source [1]	Universal positive closure architecture, exact neutral elimination, spectral gate, and LPCP.
Prior BFG sources [5,6]	Local order, fixed-point mediation, generativity/return architecture, and cross-scale compatibility.
Prior carrier-specific BFG sources [12,18,26,27,35,44]	Carrier-level structural constraints for atomic, biological, organismic, cognitive, phenomenal, and thermodynamic readouts.
This paper	Strong Universal Reclosure Equation, totalized admissible category, endogenous reciprocal dual-order construction, freedom-closed natural-reduction protocol, and the exact Compact Canonical Unified V4 neutral-contrast representation $\mathbb{U}[X] = \mathcal{R}_X(S_{XD}[X])$.

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Declarations

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Author contribution. Conceptualization, BFG unification architecture, theoretical formulation, interpretation, and manuscript responsibility: Marcel Theodor Wendé. The Level-0 and recursive conjugate-closure source lines used in the manuscript are credited through the numbered bibliography.

Data availability. No external empirical dataset was analyzed. Numerical results are synthetic deterministic or seeded calculations from equations and parameters fully stated in the manuscript.

Code availability. The simulation equations, parameter ranges, random seed, software versions, and deterministic initial conditions required to reproduce the displayed figures are provided in Appendix B. A public repository release may be attached to a later archival version.

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